

Supplemental Material for Quantum Change Interval: Exact Asymptotics for Minimum Error Localization

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S1 Conventions and exact excitation compression

Let

$$\mathcal{I}_n = \{[a, b] : 1 \leq a \leq b \leq n\}, \quad M_n = |\mathcal{I}_n| = \frac{n(n+1)}{2}, \quad (\text{S1.1})$$

with physical prior $1/M_n$. For an overlap $0 \leq c_n \leq 1$, choose the phase convention

$$|\psi_n\rangle = c_n|0\rangle + s_n|1\rangle, \quad s_n = \sqrt{1 - c_n^2}. \quad (\text{S1.2})$$

The interval states have Gram matrix

$$(G_n)_{I,J} = c_n^{|I \triangle J|}. \quad (\text{S1.3})$$

We write

$$L_n = \log n, \quad A(t) = 1 + \frac{2\sqrt{t}}{\pi} + \frac{\sqrt{2}t}{\pi^2}, \quad F(t) = A(t)^2. \quad (\text{S1.4})$$

For a generic argument $0 \leq x < 1$, define

$$s(x) = \sqrt{1 - x^2}, \quad J(x) = \frac{1}{2\pi} \int_0^{2\pi} \frac{d\theta}{|1 - xe^{i\theta}|}, \quad \ell(x) = \pi J(x),$$

and

$$p_1(x) = (1 - x^2)J(x)^2 = \frac{1 - x^2}{\pi^2} \ell(x)^2, \quad p_1(1) = 0.$$

For a moving overlap, $s_n = s(c_n)$ as in (S1.2), and we put

$$\ell_n = \ell(c_n), \quad \lambda_n = n(1 - c_n), \quad \tau_n = \lambda_n L_n^2, \quad h_n = np_1(c_n).$$

For a compact critical parameter $t \geq 0$, set

$$c_n(t) = 1 - \frac{t}{nL_n^2}. \quad (\text{S1.5})$$

Thus $L_n = \log n$ remains the logarithmic dimension variable, whereas ℓ_n is the elliptic integral scale. The functions J and ℓ are used only for $c < 1$; the endpoint $c = 1$ is always handled directly.

For a full M_n -label Gram matrix, P_{tr} , P_{SRM} , and P_{opt} denote the trace benchmark, the square root measurement success probability, and the Bayes optimum with external prior $1/M_n$. For $k \in \{0, 1, \geq 2\}$, $P_{\text{opt}}^{(k)}$ is the optimum for the subnormalized vectors in excitation sector k , retaining the same external prior. Singleton interval labels therefore remain present even when their vectors vanish in the strict hull sector.

Sections S4 and S5 establish the global squeeze in the compact window by combining a trace lower bound with sectorwise dual upper bounds. Section S6 extends the squeeze to moving overlaps and proves the unified hull dominance law by weighted hull compression and adaptive endpoint blocks. Section S7 provides an optional labelwise SRM refinement. Finally, Section S8 records the finite size numerical diagnostics.

Regime and main-to-supplement dependency maps

The regime map below fixes the scope of each asymptotic argument. Every row uses the physical Gram matrix (S1.3), the same M_n , and the same external uniform prior.

Regime	Scale assumptions	Proof input	Common conclusion
Ultracritical	$\tau_n = n(1 - c_n)(\log n)^2$ in a fixed compact window	global trace theorem and sector bounds	$M_n P_X = A(\tau_n)^2 + o(1)$
Inner hull dominance	$\lambda_n \rightarrow 0, h_n \rightarrow \infty$	inner bridge and Proposition S6.1	$P_X \sim p_1(c_n)^2$
Continuum	λ_n in a fixed compact subset of $(0, \infty)$	weighted cells and a squeeze uniform on compact parameter intervals	$P_X \sim p_1(c_n)^2$, with coefficient $2\lambda_n^2/\pi^4$ after $M_n/(\log n)^4$ scaling
Moving outer	$c_n \rightarrow 1, c_n < 1$ eventually, $\lambda_n \rightarrow \infty$	adaptive blocks and weighted block and tail certificates	$P_X \sim p_1(c_n)^2$
Unified	$c_n \rightarrow 1, c_n < 1$ eventually, $h_n \rightarrow \infty$, with arbitrary oscillation	subsequence analysis over the three possible limits of λ_n	$P_X \sim p_1(c_n)^2$

On the canonical fixed τ scale, h_n remains bounded; the unified hull dominance theorem instead assumes $h_n \rightarrow \infty$.

Main text conclusion	Supplement dependency	Role
Local diagonal and retained family transfer	Section S2	functional calculus and retained mass details
Compact critical window law	Sections S4–S5, including Theorem S4.2 and Proposition S5.9	trace lower bound and Bayes dual upper bound
Hull dominance law	Section S6, culminating in Theorem S6.7	inner, continuum, moving outer, and oscillatory sequences
Optional labelwise SRM refinement	Section S7, Theorem S7.1	mean square control of individual square root diagonals
Numerical evidence boundary	Section S8	postprocessed floating point diagnostics and working precision residuals

Lemma S1.1 (Elliptic endpoint scale). *As $c \uparrow 1$,*

$$\ell(c) = \log \frac{8}{1-c} + o(1). \quad (\text{S1.6})$$

Moreover, for every compact $C \Subset (0, \infty)$,

$$\sup_{\lambda \in C} |\ell(1 - \lambda/n) - \log n - \log(8/\lambda)| \rightarrow 0. \quad (\text{S1.7})$$

Proof. With

$$m = \frac{4c}{(1+c)^2}, \quad \delta = \sqrt{1-m} = \frac{1-c}{1+c},$$

the substitution $\theta = 2\phi$ gives

$$\ell(c) = \frac{2}{1+c} K(m),$$

where K uses the parameter convention of the main text. Its endpoint expansion is

$$K(m) = \log \frac{4}{\delta} + O\left(\delta^2 \log \frac{1}{\delta}\right).$$

Substitution of $\delta = (1 - c)/(1 + c)$ proves (S1.6). If $c = 1 - \lambda/n$ and λ lies in a compact subset of $(0, \infty)$, the same remainder is uniform in λ , which proves (S1.7). $\square$

When $c_n > 0$, put $q_n = s_n/c_n$ and $(D_n)_{I,I} = c_n^{|I|}$. The state matrix, with rows indexed by interval labels and columns by excitation subsets, is

$$(A_n)_{I,S} = c_n^{|I|} q_n^{|S|} \mathbf{1}_{\{S \subseteq I\}}, \quad G_n = A_n A_n^*. \quad (\text{S1.8})$$

Thus $\text{tr} \sqrt{G_n} = \|A_n\|_{S_1}$. In every bounded critical window, $c_n \rightarrow 1$; hence $c_n = 0$ can occur only in a finite prefix. Indices with $c_n = 1$ are handled directly, since $G_n = J_{M_n}$ and $P_{\text{tr}} = P_{\text{SRM}} = P_{\text{opt}} = M_n^{-1}$.

Proposition S1.2 (Exact physical hull coordinates). *Let Z_n be the proper-interval containment matrix*

$$(Z_n)_{I,(s,t)} = \mathbf{1}_{\{s < t, [s,t] \subseteq I\}}. \quad (\text{S1.9})$$

For $0 < c < 1$, set

$$|e_c\rangle = \frac{|\psi\rangle - c|0\rangle}{s(c)}$$

and, for $1 \leq u < v \leq n$,

$$|\eta_{u,v}\rangle = |0\rangle^{\otimes(u-1)} |e_c\rangle_u |\psi\rangle^{\otimes(v-u-1)} |e_c\rangle_v |0\rangle^{\otimes(n-v)}.$$

These vectors form an orthonormal family, and the exact coordinate state matrix of the depth-at-least-two sector is

$$[A_{\geq 2,n}(c)]_{(a,b),(u,v)} = (1 - c^2) c^{(u-a)+(b-v)} \mathbf{1}_{\{a \leq u < v \leq b\}}. \quad (\text{S1.10})$$

In particular, all M_n physical rows are retained, while the column set has cardinality $n(n-1)/2$. Define

$$[V_n(c)]_{a,u} = c^{u-a} \mathbf{1}_{\{a \leq u\}},$$

and let $R_{\mathcal{I}}$ restrict ordered row pairs to $a \leq b$ and $C_{<}$ include strict column pairs $u < v$. Then

$$A_{\geq 2,n}(c) = (1 - c^2) R_{\mathcal{I}} [V_n(c) \otimes V_n(c)^{\top}] C_{<}.$$

Equivalently, with

$$u_{a,c}^+(x) = s(c) c^{x-a} \mathbf{1}_{\{x \geq a\}}, \quad u_{b,c}^-(y) = s(c) c^{b-y} \mathbf{1}_{\{y \leq b\}},$$

the row belonging to $I = [a, b]$ is $P_{\{x < y\}}(u_{a,c}^+ \otimes u_{b,c}^-)$.

Corollary S1.3 (Weighted hull comparison). *For the moving critical notation $0 < c_n < 1$, the same matrix admits the exact factorization*

$$A_{\geq 2,n} = q_n^2 D_n Z_n W_n, \quad (W_n)_{(s,t),(s,t)} = c_n^{-(t-s-1)}. \quad (\text{S1.11})$$

Consequently,

$$I \preceq W_n \preceq \rho_n I, \quad \rho_n = c_n^{-(n-2)}. \quad (\text{S1.12})$$

Define the comparison Gram matrix by

$$\tilde{G}_n = D_n (J_{M_n} + q_n^2 B_{1,n} B_{1,n}^* + q_n^4 Z_n Z_n^*) D_n, \quad (\text{S1.13})$$

where $[B_{1,n}]_{I,x} = \mathbf{1}_{\{x \in I\}}$ is the interval-site incidence matrix. This retains the physical length factor D_n while replacing the strict-hull depth weights by one. Then

$$\tilde{G}_n \preceq G_n \preceq \rho_n^2 \tilde{G}_n. \quad (\text{S1.14})$$

Proof of Proposition S1.2 and Corollary S1.3. First suppose $0 < c < 1$. Expanding $|\psi\rangle = c|0\rangle + s(c)|e_c\rangle$, every excitation subset $S \subseteq [a, b]$ with $|S| \geq 2$ has unique first and last sites $u = \min S < v = \max S$. Write $S = \{u, v\} \cup T$, where $T \subseteq \{u+1, \dots, v-1\}$. For a fixed hull $u < v$, summing the unrestricted interior sites back to $|\psi\rangle^{\otimes(v-u-1)}$ leaves the scalar

$$s(c)^2 c^{[a,b]-(v-u-1)-2} = s(c)^2 c^{(u-a)+(b-v)}.$$

This establishes (S1.10) and also shows that the whole projected state lies in the displayed hull span. If two strict pairs have different first sites, the smaller first site carries $|e_c\rangle$ in one vector and $|0\rangle$ in the other. If their first sites agree but their last sites differ, the larger last site gives the same $|e_c\rangle$ -versus- $|0\rangle$ distinction. Hence the hull vectors are orthonormal. The projected tensor formula now follows entrywise from

$$[V_n(c)]_{a,u} [V_n(c)]_{v,b} = c^{(u-a)+(b-v)} \mathbf{1}_{\{a \leq u\}} \mathbf{1}_{\{v \leq b\}}.$$

For $c = c_n \in (0, 1)$, direct simplification gives

$$q_n^2 c_n^{|I|} c_n^{-(v-u-1)} = s_n^2 c_n^{(u-a)+(b-v)}.$$

Hence (S1.11) is the exact entrywise representation of the physical hull sector in (S1.10). The bounds (S1.12) and (S1.14) then follow as before, because the vacuum and one-excitation contributions are unchanged.

For a complex original overlap $\langle 0|\psi\rangle = ce^{i\varphi}$, replacing $|\psi\rangle$ by $e^{-i\varphi}|\psi\rangle$ multiplies the interval vector I by $e^{-i|I|\varphi}$. The Gram matrix changes only by diagonal-unitary congruence, and the physical rank-one hypotheses are unchanged. This justifies the nonnegative phase gauge used above.

At $c = 1$, every hypothesis is the vacuum and the hull sector vanishes, so neither $|e_c\rangle$ nor q_n is needed. At $c = 0$, set $|e_0\rangle = |\psi\rangle$. A proper interval then has the single hull column $(u, v) = (a, b)$, while singleton rows vanish; equation (S1.10) remains valid with $0^0 = 1$ at zero boundary depth. In both endpoint cases, all M_n physical labels, including singleton intervals, retain the external prior $1/M_n$. $\square$

Lemma S1.4 (Subnormalized PGM and dual bounds). *Consider M possibly subnormalized vectors under the uniform prior. Let X be their Gram matrix, and define*

$$P_{\text{PGM}}(v) := \frac{1}{M} \sum_{j=1}^M (\sqrt{X})_{jj}^2.$$

Then

$$\left(\frac{\text{tr } \sqrt{X}}{M} \right)^2 \leq P_{\text{PGM}}(v) \leq P_{\text{opt}}(v). \quad (\text{S1.15})$$

Moreover, if $K \succeq |v_j\rangle\langle v_j|$ for all j , then $MP_{\text{opt}}(v) \leq \text{tr } K$.

Proof. The polar measurement gives $P_{\text{PGM}} = M^{-1} \sum_j (\sqrt{X})_{jj}^2$, including singular X by support restriction. Cauchy-Schwarz proves the first inequality. The operator K/M satisfies every Bayes dual constraint under the uniform prior, proving the last assertion. $\square$

Lemma S1.5 (Orthogonal sector Minkowski bound). *Suppose $|\phi_j\rangle = \bigoplus_{\ell=1}^s |\phi_{\ell j}\rangle$, and let $P_{\text{opt}}^{(\ell)}$ be the optimum under the uniform prior for the subnormalized sector vectors $\{|\phi_{\ell j}\rangle\}_{j=1}^M$. Then*

$$\sqrt{P_{\text{opt}}} \leq \sum_{\ell=1}^s \sqrt{P_{\text{opt}}^{(\ell)}}. \quad (\text{S1.16})$$

Proof. For a full POVM $\{M_j\}$, set $x_{\ell j} = M^{-1/2} M_j^{1/2} \phi_{\ell j}$. Minkowski's inequality in the direct sum over labels gives

$$\left(\sum_j \left\| \sum_{\ell} x_{\ell j} \right\|^2 \right)^{1/2} \leq \sum_{\ell} \left(\sum_j \|x_{\ell j}\|^2 \right)^{1/2}.$$

Compressing the POVM to the ℓ th physical sector makes the corresponding term at most $\sqrt{P_{\text{opt}}^{(\ell)}}$. Optimizing the left-hand side proves the claim. $\square$

S2 Følner and transfer proofs

This section states and proves the functional calculus, Gram stability, dual, and retained family transfer tools used by the main text.

S2.1 Local diagonal Følner limit

Lemma S2.1 (Local diagonal Følner limit). *Let $k \in \ell^1(\mathbb{Z}^d)$ satisfy $k(-u) = \overline{k(u)}$, let A be convolution by k on $\ell^2(\mathbb{Z}^d)$, and suppose its Fourier symbol satisfies*

$$0 < \mu \leq F(\theta) \leq L < \infty$$

almost everywhere. If $\Lambda_n \subset \mathbb{Z}^d$ is any finite Følner sequence and $A_n = P_n A P_n|_{\ell^2(\Lambda_n)}$, then for every continuous $\phi : [\mu, L] \rightarrow \mathbb{C}$,

$$\frac{1}{|\Lambda_n|} \sum_{x \in \Lambda_n} |\langle \delta_x, \phi(A_n) \delta_x \rangle - a_\phi| \longrightarrow 0, \quad (\text{S2.1})$$

where

$$a_\phi = \frac{1}{(2\pi)^d} \int_{[0, 2\pi]^d} \phi(F(\theta)) d\theta.$$

Proof. Since A has spectrum in $[\mu, L]$, the same is true of every compression A_n . Fix $\varepsilon > 0$ and put $\eta = \varepsilon/8$. By the Weierstrass theorem, applied to the real and imaginary parts if necessary, choose a polynomial

$$p(t) = \sum_{m=0}^s a_m t^m \quad \text{such that} \quad \sup_{t \in [\mu, L]} |\phi(t) - p(t)| < \eta.$$

Let $k_R = k \mathbf{1}_{\{\|u\|_\infty \leq R\}}$, let A_R be convolution by k_R , and set $A_{R,n} = P_n A_R P_n|_{\ell^2(\Lambda_n)}$. Young's inequality gives

$$\|A - A_R\|_{\text{op}}, \|A_n - A_{R,n}\|_{\text{op}} \leq \|k - k_R\|_{\ell^1}. \quad (\text{S2.2})$$

Choose R so large that the right-hand side of (S2.2) is at most 1 and

$$C_p \|k - k_R\|_{\ell^1} < \eta, \quad C_p := \sum_{m=1}^s m |a_m| (L+1)^{m-1},$$

where the second condition is void if $C_p = 0$. The telescoping identity for $X^m - Y^m$, together with $\|A\|, \|A_n\| \leq L$ and $\|A_R\|, \|A_{R,n}\| \leq L+1$, then yields

$$\|p(A) - p(A_R)\|_{\text{op}}, \|p(A_n) - p(A_{R,n})\|_{\text{op}} < \eta. \quad (\text{S2.3})$$

Define

$$Q_t = \{h \in \mathbb{Z}^d : \|h\|_\infty \leq t\}$$

and the sR -interior and boundary by

$$\begin{aligned}\Lambda_n^{\circ, sR} &= \{x \in \Lambda_n : x + Q_{sR} \subseteq \Lambda_n\}, \\ \partial_{sR}\Lambda_n &= \Lambda_n \setminus \Lambda_n^{\circ, sR}.\end{aligned}$$

Because a polynomial of degree s in an operator of propagation range R has propagation radius sR , every path of length $m \leq s$ generated by the kernel stays in Λ_n when it starts at $x \in \Lambda_n^{\circ, sR}$. Therefore

$$\langle \delta_x, p(A_{R,n})\delta_x \rangle = \langle \delta_0, p(A_R)\delta_0 \rangle, \quad x \in \Lambda_n^{\circ, sR}. \quad (\text{S2.4})$$

Moreover,

$$\partial_{sR}\Lambda_n \subseteq \bigcup_{\|h\|_\infty \leq sR} (\Lambda_n \setminus (\Lambda_n - h)). \quad (\text{S2.5})$$

The index set in this union is finite. Each set has $o(|\Lambda_n|)$ elements by the Følner property, and hence $|\partial_{sR}\Lambda_n|/|\Lambda_n| \rightarrow 0$. With $K_p = \sup_{|t| \leq L+1} |p(t)|$, define

$$\begin{aligned}d_{n,R,p}(x) &:= |\langle \delta_x, p(A_{R,n})\delta_x \rangle - \langle \delta_0, p(A_R)\delta_0 \rangle|, \\ D_{n,R,p} &:= \frac{1}{|\Lambda_n|} \sum_{x \in \Lambda_n} d_{n,R,p}(x).\end{aligned}$$

Equations (S2.4)–(S2.5) give

$$D_{n,R,p} \leq 2K_p \frac{|\partial_{sR}\Lambda_n|}{|\Lambda_n|} \rightarrow 0. \quad (\text{S2.6})$$

Finally, the spectral theorem and Fourier transformation identify

$$\langle \delta_0, \phi(A)\delta_0 \rangle = \frac{1}{(2\pi)^d} \int_{[0, 2\pi]^d} \phi(F(\theta)) d\theta = a_\phi.$$

Put

$$D_n^\phi := \frac{1}{|\Lambda_n|} \sum_{x \in \Lambda_n} |\langle \delta_x, \phi(A_n)\delta_x \rangle - a_\phi|.$$

For all sufficiently large n , functional calculus, (S2.3), and (S2.6) yield

$$\begin{aligned}D_n^\phi &\leq \|\phi(A_n) - p(A_n)\|_{\text{op}} \\ &\quad + \|p(A_n) - p(A_{R,n})\|_{\text{op}} \\ &\quad + D_{n,R,p} \\ &\quad + \|p(A_R) - p(A)\|_{\text{op}} \\ &\quad + \|p(A) - \phi(A)\|_{\text{op}} \\ &< 4\eta + \frac{\varepsilon}{2} = \varepsilon.\end{aligned}$$

This proves the claimed local diagonal limit. $\square$

S2.2 Gram stability and dual control

Lemma S2.2 (Operator norm Gram stability). *Let $G, H \in \mathbb{C}^{M \times M}$ be positive semidefinite with unit diagonal. Then*

$$|P_{\text{opt}}(G) - P_{\text{opt}}(H)| \leq \|\sqrt{G} - \sqrt{H}\|_{\text{op}} \leq \|G - H\|_{\text{op}}^{1/2} \quad (\text{S2.7})$$

and

$$\begin{aligned} & |P_{\text{SRM}}(G) - P_{\text{SRM}}(H)| \\ & \leq 2\|\sqrt{G} - \sqrt{H}\|_{\text{op}} \leq 2\|G - H\|_{\text{op}}^{1/2}. \end{aligned} \quad (\text{S2.8})$$

If $G, H \succeq \mu I$ for some $\mu > 0$, then

$$\|\sqrt{G} - \sqrt{H}\|_{\text{op}} \leq \frac{\|G - H\|_{\text{op}}}{2\sqrt{\mu}}. \quad (\text{S2.9})$$

Proof. Use the canonical Gram realizations $|x_j\rangle = \sqrt{G}e_j$ and $|y_j\rangle = \sqrt{H}e_j$. Both vectors have unit norm because the Gram matrices have unit diagonal, and the trace distance between the corresponding pure states is at most $\|x_j - y_j\|$. Evaluate an optimal POVM for either ensemble on the other and average over j to obtain the first inequality in (S2.7). The second is the operator Hölder inequality for positive square roots. Since $0 \leq (\sqrt{G})_{jj}, (\sqrt{H})_{jj} \leq 1$, the scalar bound $|a^2 - b^2| \leq 2|a - b|$ gives (S2.8). To prove (S2.9), $X = \sqrt{G} - \sqrt{H}$ solves

$$\sqrt{G}X + X\sqrt{H} = G - H.$$

The inverse Sylvester operator has norm at most $(2\sqrt{\mu})^{-1}$, which proves the claim. $\square$

Lemma S2.3 (Gram dual bound). *Let G be an $M \times M$ positive semidefinite Gram matrix with unit diagonal. Set*

$$\begin{aligned} X &= \sqrt{G}, & d_j &= X_{jj}, \\ \bar{d} &= \frac{\text{tr } X}{M}, & |x_j\rangle &= X|e_j\rangle. \end{aligned}$$

Then $\bar{d} > 0$, and the operator

$$Y_G = \frac{1}{M} \left[\bar{d}X + \sum_{j:d_j > \bar{d}} \left(1 - \frac{\bar{d}}{d_j}\right) |x_j\rangle\langle x_j| \right] \quad (\text{S2.10})$$

is feasible for the minimum error dual problem. Consequently,

$$\begin{aligned} \bar{d}^2 &\leq P_{\text{SRM}}(G) \leq P_{\text{opt}}(G) \\ &\leq \text{tr } Y_G \\ &= \bar{d}^2 + \frac{1}{M} \sum_{j:d_j > \bar{d}} \left(1 - \frac{\bar{d}}{d_j}\right) \\ &\leq \bar{d}^2 + \frac{1}{2M\bar{d}} \sum_{j=1}^M |d_j - \bar{d}|. \end{aligned} \quad (\text{S2.11})$$

Proof. The lower bound follows from (S1.15). Since $\text{tr } G = M > 0$, also $\bar{d} > 0$. Put $\rho_j = |x_j\rangle\langle x_j|$. For every vector $|v\rangle$, Cauchy–Schwarz gives

$$\begin{aligned} \langle v|\rho_j|v\rangle &= |\langle e_j|Xv\rangle|^2 \\ &= |\langle G^{1/4}e_j|G^{1/4}v\rangle|^2 \\ &\leq d_j \langle v|X|v\rangle. \end{aligned}$$

Thus $\rho_j \preceq d_j X$, including when G is singular. If $d_j \leq \bar{d}$, the first term in MY_G already dominates ρ_j . If $d_j > \bar{d}$, then

$$\bar{d}X \succeq \frac{\bar{d}}{d_j} \rho_j,$$

and the j th correction in (S2.10) supplies the remaining multiple of ρ_j . All other corrections are positive semidefinite. Hence $Y_G \succeq \rho_j/M$ for every j , so Y_G is dual feasible. Finally, $\text{tr } \rho_j = G_{jj} = 1$, and for $d_j > \bar{d}$,

$$1 - \frac{\bar{d}}{d_j} = \frac{d_j - \bar{d}}{d_j} \leq \frac{|d_j - \bar{d}|}{\bar{d}}.$$

The Bayes dual constraints under the uniform prior are $Y \succeq \rho_j/M$, so weak duality gives $P_{\text{opt}}(G) \leq \text{tr } Y_G$. Finally, $\sum_j (d_j - \bar{d}) = 0$, so the sum of the positive deviations is one half of $\sum_j |d_j - \bar{d}|$. Taking the trace of (S2.10) proves the remaining bounds. $\square$

S2.3 Retained family Gram transfer

Proposition S2.4 (Retained family Gram transfer). *For each n , let G_n be an $M_n \times M_n$ positive semidefinite Gram matrix with unit diagonal. Let $\mathcal{B}_n$ be a set of m_n labels, let $G_{n,\mathcal{B}}$ be the associated principal submatrix, and let H_n be an $m_n \times m_n$ positive semidefinite matrix with unit diagonal. Suppose*

$$\frac{m_n}{M_n} \rightarrow 1, \tag{S2.12}$$

$$\|G_{n,\mathcal{B}} - H_n\|_{\text{op}} \rightarrow 0, \tag{S2.13}$$

and, for some $\alpha > 0$,

$$\frac{1}{m_n} \sum_{j=1}^{m_n} \left| (\sqrt{H_n})_{jj} - \alpha \right| \rightarrow 0. \tag{S2.14}$$

Then the complete ensemble under the uniform prior satisfies

$$P_{\text{opt}}(G_n), P_{\text{SRM}}(G_n) \rightarrow \alpha^2. \tag{S2.15}$$

Proof. Put $d_{n,j} = (\sqrt{H_n})_{jj}$ and $\bar{d}_n = m_n^{-1} \sum_j d_{n,j}$. Positivity and the unit diagonal give $0 \leq d_{n,j} \leq 1$, because

$$d_{n,j}^2 \leq \langle e_j, H_n e_j \rangle = 1.$$

The local-diagonal hypothesis implies $\bar{d}_n \rightarrow \alpha$ and

$$\frac{1}{m_n} \sum_j d_{n,j}^2 \rightarrow \alpha^2.$$

The latter is $P_{\text{SRM}}(H_n)$. Put

$$a_n = \frac{1}{m_n} \sum_j |d_{n,j} - \alpha|.$$

Then $|\bar{d}_n - \alpha| \leq a_n \rightarrow 0$, so $\bar{d}_n \geq \alpha/2$ for all sufficiently large n , and

$$\frac{1}{2m_n \bar{d}_n} \sum_j |d_{n,j} - \bar{d}_n| \leq \frac{a_n}{\bar{d}_n} \rightarrow 0.$$

Lemma S2.3 now gives

$$P_{\text{opt}}(H_n) \rightarrow \alpha^2.$$

By Lemma S2.2 and (S2.13), the retained physical family has the same limits:

$$P_{\text{opt}}(G_{n,\mathcal{B}}), P_{\text{SRM}}(G_{n,\mathcal{B}}) \longrightarrow \alpha^2. \quad (\text{S2.16})$$

The same square root perturbation bound also yields

$$\frac{\text{tr} \sqrt{G_{n,\mathcal{B}}}}{m_n} \longrightarrow \alpha. \quad (\text{S2.17})$$

Write $\beta_n = m_n/M_n$. Restricting a full ensemble POVM to the retained labels and completing the resulting sub-POVM gives the upper bound below; extending an optimal retained family POVM gives the lower bound:

$$\beta_n P_{\text{opt}}(G_{n,\mathcal{B}}) \leq P_{\text{opt}}(G_n) \leq \beta_n P_{\text{opt}}(G_{n,\mathcal{B}}) + (1 - \beta_n). \quad (\text{S2.18})$$

The retained-mass hypothesis, (S2.16), and (S2.18) prove $P_{\text{opt}}(G_n) \rightarrow \alpha^2$. It remains to transfer the SRM. Put

$$t_n = \frac{\text{tr} \sqrt{G_n}}{M_n}.$$

Eigenvalue interlacing for the principal submatrix gives

$$t_n \geq \beta_n \frac{\text{tr} \sqrt{G_{n,\mathcal{B}}}}{m_n} \longrightarrow \alpha.$$

On the other hand, $t_n^2 \leq P_{\text{opt}}(G_n) \rightarrow \alpha^2$, so $t_n \rightarrow \alpha$. The universal squeeze

$$t_n^2 \leq P_{\text{SRM}}(G_n) \leq P_{\text{opt}}(G_n)$$

proves the stated SRM conclusion. $\square$

Corollary S2.5 (Følner comparison transfer). *Let $k \in \ell^1(\mathbb{Z}^d)$ be Hermitian with $k(0) = 1$, and suppose the symbol F of its convolution operator satisfies $0 < \mu \leq F \leq L < \infty$. Let Λ_n be a finite Følner sequence and H_n the compression of this convolution operator to $\ell^2(\Lambda_n)$. If G_n and retained label sets $\mathcal{B}_n$ satisfy*

$$\frac{|\mathcal{B}_n|}{M_n} \rightarrow 1, \quad \|G_n[\mathcal{B}_n] - H_n\|_{\text{op}} \rightarrow 0,$$

define

$$\alpha_F = \frac{1}{(2\pi)^d} \int_{[0,2\pi]^d} \sqrt{F(\theta)} d\theta. \quad (\text{S2.19})$$

Then

$$P_{\text{opt}}(G_n), P_{\text{SRM}}(G_n) \longrightarrow \alpha_F^2. \quad (\text{S2.20})$$

Proof. The matrices H_n have unit diagonal. Lemma S2.1, applied to $\phi(t) = \sqrt{t}$, gives (S2.14) with $\alpha = \alpha_F > 0$. Proposition S2.4 applies. $\square$

S3 Retained orthogonality

The global trace argument and the local refinement use the same retained subspaces, which we now construct.

Define the unweighted three-stratum comparison

$$\widehat{G}_n = J_{M_n} + q_n^2 B_{1,n} B_{1,n}^* + q_n^4 Z_n Z_n^*, \quad (\text{S3.1})$$

where Z_n retains its zero singleton rows. Let $z_n = B_{1,n}^* \mathbf{1}$, and let $Q_{1,n}$ project onto $z_n^\perp$. Then $\text{ran}(B_{1,n} Q_{1,n}) \perp \mathbf{1}$.

Partition the sites into $r_n = \lfloor \sqrt{\log(n+1)} \rfloor$ balanced consecutive blocks $E_1, \dots, E_{r_n}$. For a left block E_p , set

$$F_p(a, s) = \mathbf{1}_{\{a \leq s\}}, \quad s \in E_p,$$

and let Q_p^L project onto the orthogonal complement of $\text{span}\{\mathbf{1}_{E_p}, F_p^* \mathbf{1}_n\}$. For a right block E_r , define $R_r(b, t) = \mathbf{1}_{\{t \leq b\}}$ and let Q_r^R project onto the orthogonal complement of $\text{span}\{\mathbf{1}_{E_r}, R_r^* \mathbf{1}_n\}$. On pair columns $E_p \times E_r$ with $p < r$, retain $Q_p^L \otimes Q_r^R$; retain zero when both sites lie in one block. Their direct sum is denoted by $Q_{2,n}$.

Lemma S3.1 (Exact retained support and row orthogonality). *For every $p < r$, the (p, r) column block of $Z_n Q_{2,n}$ is supported on interval rows whose endpoint pair belongs to $E_p \times E_r$. On that cell it factorizes, up to reversal of the right coordinate, as*

$$(U_{|E_p|} Q_p^L) \otimes (U_{|E_r|} Q_r^R). \quad (\text{S3.2})$$

Different cells have orthogonal row and column supports, and the full retained row range is orthogonal to every endpoint-additive function $f(a) + g(b)$.

Proof. If a lies before E_p , its row in F_p is constant and is killed by Q_p^L ; if it lies after E_p , that row is zero. Thus $F_p Q_p^L$ is supported on $a \in E_p$. Moreover, $\mathbf{1}_n^* F_p Q_p^L = 0$ by the definition of Q_p^L . The analogous statements hold for $R_r Q_r^R$. Since $p < r$, the Cartesian support cell lies entirely in $a < b$, where containment factorizes, proving (S3.2). Every tensor output has zero sum in each endpoint coordinate, so it is orthogonal to $f(a) + g(b)$. Distinct block pairs have disjoint row and column supports. $\square$

S4 Incidence trace laws and the global trace theorem

For the interval-site incidence matrix $B_{1,n}$ defined above, a direct count gives

$$B_{1,n}^* B_{1,n} = (n+1) L_D^{-1}, \quad (\text{S4.1})$$

where L_D is the Dirichlet path Laplacian. Therefore

$$\sigma_k(B_{1,n}) = \frac{\sqrt{n+1}}{2 \sin[\pi k / (2(n+1))]}, \quad 1 \leq k \leq n, \quad (\text{S4.2})$$

and endpoint harmonic summation yields

$$\|B_{1,n}\|_{S_1} = \frac{n^{3/2}}{\pi} \log n + O(n^{3/2}), \quad \|B_{1,n}\|_{\text{op}} = O(n^{3/2}). \quad (\text{S4.3})$$

The corresponding two-endpoint incidence operator is compressed to a triangular domain rather than a square one.

Proposition S4.1 (Triangular Volterra trace law). *For the matrix Z_n in (S1.9),*

$$\|Z_n\|_{S_1} \sim \frac{n^2 (\log n)^2}{2\pi^2}. \quad (\text{S4.4})$$

More quantitatively, for an integer $2 \leq R \leq n$, partition the sites into R balanced consecutive blocks of sizes ℓ_p . Retain only matched row and column cells with $p < r$, and project each one-dimensional column factor off the constant and incidence-sum directions used in Lemma S3.1. The resulting rectangular pinching $Z_{n,R}^{\text{ret}}$ satisfies

$$\|Z_{n,R}^{\text{ret}}\|_{S_1} \geq \frac{n^2 (\log n)^2}{2\pi^2} \left[1 - C \left(\frac{1}{R} + \frac{\log R}{\log n} + \frac{1}{\log n} + \frac{R}{n} \right) \right]. \quad (\text{S4.5})$$

Proof. For $U_m(i, j) = \mathbf{1}_{\{i \leq j\}}$,

$$\|U_m\|_{S_1} = \frac{m}{\pi} \log m + O(m). \quad (\text{S4.6})$$

On a matched off-diagonal cell (p, r) , the unprojected block is $U_{\ell_p} \otimes U_{\ell_r}^\top$. Each projection removes at most two column directions, and therefore

$$\|U_\ell(I - Q)\|_{S_1} \leq \sqrt{2} \|U_\ell\|_{S_2} = O(\ell).$$

Thus each one-dimensional trace norm changes by at most $O(\ell_p)$ or $O(\ell_r)$. The retained row and column cells are mutually orthogonal, so rectangular pinching and (S4.6) give

$$\|Z_{n,R}^{\text{ret}}\|_{S_1} \geq \frac{1}{\pi^2} \sum_{p < r} \ell_p \ell_r (\log \ell_p - C)(\log \ell_r - C).$$

The identity $\sum_{p < r} \ell_p \ell_r = (n^2 - \sum_p \ell_p^2)/2$, together with the balance of the block sizes, proves (S4.5).

For the full upper bound, fix an integer $R \geq 2$ and decompose both endpoint triangles into all macroblocks. Blocks with two matching endpoint indices are compressions of $U \otimes U$; blocks with one matching index are compressions of $U \otimes J$; and blocks with no matching index have rank one. For a balanced side length ℓ ,

$$\|U_\ell \otimes J_\ell\|_{S_1} = O(\ell^2 \log \ell), \quad \|J_\ell \otimes J_\ell\|_{S_1} = O(\ell^2).$$

Thus, after summing the finitely many macroblocks for fixed R , the blocks with one or zero matching endpoint indices contribute $o(n^2 \log^2 n)$. The off-diagonal cells with two matching endpoint indices have total normalized contribution

$$\frac{1}{\pi^2 n^2 \log^2 n} \sum_{p < r} \ell_p \ell_r \log \ell_p \log \ell_r \longrightarrow \frac{1 - R^{-1}}{2\pi^2}.$$

The R diagonal cells are rescaled triangular problems. Their total limsup contribution is at most C^*/R , because $R(n/R)^2 \log^2(n/R)/[n^2 \log^2 n] \rightarrow R^{-1}$. To justify the use of the finite constant C^* , note first that, up to the fixed reversal of the right-endpoint coordinate,

$$\Lambda_m = \{(x, y) \in \{1, \dots, m\}^2 : x + y \leq m + 1\}, \quad Z_n = P_{\Lambda_{n-1}}(U_{n-1} \otimes U_{n-1})P_{\Lambda_{n-1}}.$$

The ideal property and (S4.6) therefore give

$$\|Z_n\|_{S_1} \leq \|U_{n-1}\|_{S_1}^2 = O(n^2 \log^2 n).$$

Hence, if $C^* = \limsup_n \|Z_n\|_{S_1}/[n^2 \log^2 n]$, the block triangle inequality gives

$$C^* \leq \frac{1 - R^{-1}}{2\pi^2} + \frac{C^*}{R}.$$

Thus $C^* \leq 1/(2\pi^2)$. The retained lower bound followed by $R \rightarrow \infty$ gives the reverse inequality and proves (S4.4). $\square$

Together, these sector estimates yield the global trace law.

Theorem S4.2 (Critical global square root trace). *Let $0 \leq \tau_n = n(1 - c_n)(\log n)^2 \leq T$. Then*

$$\frac{\text{tr} \sqrt{G_n}}{\sqrt{M_n}} \geq A(\tau_n) - C_T \frac{\log \log n}{\log n}, \quad (\text{S4.7})$$

and, uniformly for $\tau_n \in [0, T]$,

$$\frac{\text{tr} \sqrt{G_n}}{\sqrt{M_n}} - A(\tau_n) \longrightarrow 0. \quad (\text{S4.8})$$

Proof. Choose $R_n = \lfloor \log n / \log \log n \rfloor$. Work first with the unweighted state matrix

$$\hat{A}_n = [\mathbf{1}, q_n B_{1,n}, q_n^2 Z_n].$$

Project the one-excitation columns by $Q_{1,n}$, and use the block projections of Lemma S3.1 in the two-endpoint sector. The three retained row ranges are mutually orthogonal, so the trace norm of their concatenation is the sum of their trace norms. The one-excitation loss is explicitly bounded by

$$\frac{q_n}{\sqrt{M_n}} (\|B_{1,n}\|_{S_1} - \|B_{1,n} Q_{1,n}\|_{S_1}) \leq \frac{q_n \|B_{1,n}\|_{\text{op}}}{\sqrt{M_n}} = O_T((\log n)^{-1}). \quad (\text{S4.9})$$

For the two-endpoint blocks, the finite-codimension bound in the proof of Proposition S4.1 is $O(\ell)$ per Volterra factor, hence a relative $O(1/\log \ell) = O(1/\log n)$ loss. Consequently, (S4.5) applies to these same projections.

Equations (S4.3) and (S4.5), together with

$$\begin{aligned} q_n \sqrt{n} \log n &= \sqrt{2\tau_n} [1 + O_T((n(\log n)^2)^{-1})], \\ q_n^2 n (\log n)^2 &= 2\tau_n [1 + O_T((n(\log n)^2)^{-1})] \end{aligned}$$

give the unweighted version of (S4.7). Indeed, the error in (S4.5) is

$$O\left(R_n^{-1} + \frac{\log R_n}{\log n} + \frac{1}{\log n} + \frac{R_n}{n}\right) = O\left(\frac{\log \log n}{\log n}\right).$$

In particular, for the unweighted comparison matrix in (S3.1),

$$\sup_{0 \leq t \leq T} \left| \frac{\text{tr} \sqrt{\hat{G}_n}}{\sqrt{M_n}} - A(t) \right| \longrightarrow 0. \quad (\text{S4.10})$$

To restore the physical weights, observe that $\|D_n - I\|_{\text{op}} = O_T((\log n)^{-2})$ and $\|W_n - I\|_{\text{op}} = O_T((\log n)^{-2})$. The ideal property gives

$$\begin{aligned} \|(D_n - I)\hat{A}_n\|_{S_1} &\leq \|D_n - I\|_{\text{op}} \|\hat{A}_n\|_{S_1}, \\ q_n^2 \|D_n Z_n (W_n - I)\|_{S_1} &\leq q_n^2 \|W_n - I\|_{\text{op}} \|Z_n\|_{S_1}. \end{aligned}$$

After division by $\sqrt{M_n}$, both errors are $O_T((\log n)^{-2})$. The trace norm is Lipschitz under these perturbations of the state matrix, so the physical matrix obeys (S4.7).

For the matching upper limit, apply the triangle inequality for the Schatten 1-norm to the vacuum, one-excitation, and exact hull sectors, then use (S4.3), (S4.4), and Corollary S1.3. This gives uniformly on compact parameter windows,

$$\limsup_{n \rightarrow \infty} \left[\frac{\text{tr} \sqrt{G_n}}{\sqrt{M_n}} - A(\tau_n) \right] \leq 0.$$

Combined with the retained lower bound, this proves (S4.8). $\square$

The sector upper bound is established by Proposition S5.9 in Section S5. Together with Theorem S4.2 and $P_{\text{tr}} \leq P_{\text{SRM}} \leq P_{\text{opt}}$, it gives the common rate on compact parameter windows. Retained rectangular pinching supplies the lower estimate, and dual feasibility supplies the upper estimate.

S5 Cyclic and Volterra dual certificates

For vectors $v_1, \dots, v_R$, define the geometric dual value

$$\kappa(v_1, \dots, v_R) = \inf\{\text{tr } K : K \succeq 0, K \succeq |v_j\rangle\langle v_j| \text{ for every } j\}. \quad (\text{S5.1})$$

The following rank one criterion also covers singular Gram matrices:

$$|v\rangle\langle v| \preceq K \iff v \in \text{ran } K \text{ and } \langle v, K^+ v \rangle \leq 1. \quad (\text{S5.2})$$

S5.1 The cyclic one-excitation certificate

Let $u_{a,L}$ be the indicator of the oriented length- L arc beginning at a on the cycle $\mathbb{Z}_n$, and set

$$\Sigma_n = \sum_{a \in \mathbb{Z}_n} \sum_{L=1}^n |u_{a,L}\rangle\langle u_{a,L}|. \quad (\text{S5.3})$$

Lemma S5.1 (Cyclic spectrum and leverage). *In the normalized Fourier basis, the eigenvalues of Σ_n are*

$$\lambda_0 = \frac{n(n+1)(2n+1)}{6}, \quad \lambda_k = \frac{n}{2 \sin^2(\pi k/n)}, \quad 1 \leq k < n. \quad (\text{S5.4})$$

Consequently,

$$\text{tr } \sqrt{\Sigma_n} = \frac{\sqrt{2}}{\pi} n^{3/2} \log n + O(n^{3/2}), \quad (\text{S5.5})$$

$$\max_{1 \leq L \leq n} \langle u_{0,L}, \Sigma_n^{-1/2} u_{0,L} \rangle \leq \frac{\sqrt{2}}{\pi} \frac{\log n}{\sqrt{n}} + O(n^{-1/2}). \quad (\text{S5.6})$$

Proof. Translation invariance diagonalizes Σ_n . For $k \neq 0$,

$$\sum_{L=1}^n \left| \sum_{x=0}^{L-1} e^{2\pi i k x/n} \right|^2 = \frac{1}{\sin^2(\pi k/n)} \sum_{L=1}^n \sin^2(\pi k L/n) = \frac{n}{2 \sin^2(\pi k/n)},$$

while the zero mode gives $\sum_L L^2$. Endpoint harmonic summation proves (S5.5). The leverage is

$$\frac{L^2}{n\sqrt{\lambda_0}} + \frac{\sqrt{2}}{n^{3/2}} \sum_{k=1}^{n-1} \frac{\sin^2(\pi k L/n)}{\sin(\pi k/n)}.$$

Use $\sin^2 x = (1 - \cos 2x)/2$, the endpoint expansion of the cosecant, and the uniform lower bound for partial cosine-harmonic sums $\inf_{N,x} \sum_{k=1}^N \cos(kx)/k > -\infty$. The sum is at most $(n/\pi) \log n + O(n)$, uniformly in L , proving (S5.6). $\square$

Let t_n denote the maximum leverage in (S5.6). By (S5.2),

$$K_n^{\text{cyc}} = t_n \sqrt{\Sigma_n} \succeq |u_{a,L}\rangle\langle u_{a,L}|, \quad \text{tr } K_n^{\text{cyc}} \leq \frac{2}{\pi^2} n (\log n)^2 + Cn \log n. \quad (\text{S5.7})$$

Every linear interval is a cyclic arc. Its physical one-excitation vector is $s_n c_n^{L-1} u_{a,L}$, whose scalar factor is at most s_n . Thus the physical one-excitation dual is $s_n^2 K_n^{\text{cyc}}$.

S5.2 Endpoint odd-cosine estimates and boundary completion

Let $r_{m,j} = \mathbf{1}_{\{j, \dots, m\}}$, let U_m have these row vectors, and set

$$S_m = U_m^\top U_m, \quad C_m = S_m^{1/2}, \quad \gamma_{m,j} = \langle r_{m,j}, S_m^{-1/2} r_{m,j} \rangle. \quad (\text{S5.8})$$

Lemma S5.2 (Odd-cosine endpoint bounds). *There is an absolute C such that*

$$\gamma_{m,j} \leq \frac{1}{\pi} \left(\log m + \log_+ \frac{m}{j} \right) + C, \quad 1 \leq j \leq m, \quad (\text{S5.9})$$

and

$$\text{tr } C_m = \frac{m}{\pi} \log m + O(m), \quad (\text{S5.10})$$

$$\sum_{x=1}^m \left| (C_m)_{xx} - \frac{\text{tr } C_m}{m} \right| = O(m). \quad (\text{S5.11})$$

Proof. Reversing coordinates turns $U_m U_m^\top$ into the Brownian covariance matrix. With

$$\theta_k = \frac{(2k-1)\pi}{2m+1}, \quad \lambda_k = \frac{1}{4 \sin^2(\theta_k/2)}, \quad v_k(p) = \frac{2}{\sqrt{2m+1}} \sin(p\theta_k),$$

the rectangular polar identity gives

$$\gamma_{m,j} = \frac{1}{2m+1} \sum_{k=1}^m \frac{1 + \cos[(2j-1)\theta_k]}{\sin(\theta_k/2)}. \quad (\text{S5.12})$$

Uniformly in k , $\csc(\theta_k/2) = 2(2m+1)/[\pi(2k-1)] + O(1)$. Put $x_j = (2j-1)\pi/(2m+1)$. Abel summation and

$$\left| \sum_{k=1}^K \cos((2k-1)x) \right| \leq \min\{K, |\sin x|^{-1}\}$$

show that the truncated odd-harmonic sum differs from its infinite series by at most

$$\frac{C}{m \sin x_j} = O(1),$$

uniformly in j . Hence the identity

$$\sum_{k \geq 1} \frac{\cos[(2k-1)x]}{2k-1} = \frac{1}{2} \log \cot(x/2), \quad 0 < x < \pi,$$

implies

$$\sum_{k=1}^m \frac{\cos[(2k-1)x_j]}{2k-1} = \frac{1}{2} \log \cot(x_j/2) + O(1).$$

Together with $\sum_{k=1}^m (2k-1)^{-1} = \frac{1}{2} \log m + O(1)$ and $\log \cot(x_j/2) \leq \log_+(Cm/j)$, this proves (S5.9), including $j=1, m$. Summing the singular values gives (S5.10).

For the physical-coordinate diagonal, the exact expansion is

$$(C_m)_{xx} = \frac{1}{2m+1} \sum_{k=1}^m \frac{1 - \cos(2x\theta_k)}{\sin(\theta_k/2)}. \quad (\text{S5.13})$$

Let $D_{xk} = \cos(2x\theta_k)$. Discrete odd-cosine orthogonality gives

$$D^\top D = \frac{2m+1}{4} I - \frac{1}{2} J. \quad (\text{S5.14})$$

Let $a_k = \csc(\theta_k/2)$, and let $d_x = (C_m)_{xx}$. Centering can only decrease the Euclidean norm, so

$$\frac{1}{m} \sum_{x=1}^m \left| d_x - \frac{1}{m} \sum_{y=1}^m d_y \right|^2 \leq \frac{a^\top D^\top D a}{m(2m+1)^2} \leq \frac{\|a\|_2^2}{4m(2m+1)} = O(1),$$

where the last step uses $\sum_k \csc^2(\theta_k/2) = O(m^2)$. Since the centered mean is $\text{tr } C_m/m$, Cauchy–Schwarz now gives (S5.11). $\square$

Proposition S5.3 (Quantitative Volterra certificate). *For all sufficiently large m , set*

$$\delta_m = \frac{3 \log \log m}{\log m}, \quad h_m = \lceil m^{1-\delta_m} \rceil. \quad (\text{S5.15})$$

There is $K_m \succeq 0$ dominating every suffix projector and satisfying

$$\text{tr } K_m \leq \frac{m(\log m)^2}{\pi^2} \left[1 + C \frac{\log \log m}{\log m} \right], \quad (\text{S5.16})$$

$$\frac{1}{\text{tr } K_m} \sum_x \left| (K_m)_{xx} - \frac{\text{tr } K_m}{m} \right| \leq \frac{C}{\log m}. \quad (\text{S5.17})$$

Proof. Let $t_m = \max_{j \geq h_m} \gamma_{m,j}$. Lemma S5.2 gives $t_m \leq (1+\delta_m) \log m/\pi + C$, and $t_m C_m$ dominates all suffixes beginning at $j \geq h_m$. For $j < h_m$, decompose

$$r_{m,j} = w_m + \ell_{m,j}, \quad w_m = \mathbf{1}_{\{h_m, \dots, m\}}, \quad \ell_{m,j} = \mathbf{1}_{\{j, \dots, h_m-1\}}.$$

The cyclic certificate (S5.7), embedded in the first $h_m - 1$ coordinates, supplies K_{h_m} with $\text{tr } K_{h_m} = O(h_m \log^2 h_m)$. Therefore

$$K_m = t_m C_m + 2(|w_m\rangle\langle w_m| + K_{h_m}) \quad (\text{S5.18})$$

dominates every suffix. The chosen boundary scale obeys

$$m^{-\delta_m} = O((\log m)^{-3}).$$

Thus the completion has relative trace cost $O((\log m)^{-2})$. Equation (S5.10) proves (S5.16). The diagonal estimate (S5.11) then proves (S5.17). Feasibility and Lemma S1.4 provide the lower estimate $\text{tr } K_m \geq c_0 m \log^2 m$, for an absolute constant $c_0 > 0$, used in the relative bound. $\square$

Corollary S5.4 (Fixed-boundary Volterra certificate). *Fix $0 < \delta < 1$ and put $h_{m,\delta} = \lceil m^{1-\delta} \rceil$. The same construction gives a certificate $K_{m,\delta}$ dominating every suffix projector and satisfying*

$$\text{tr } K_{m,\delta} \leq \left[\frac{1+\delta}{\pi^2} + o_\delta(1) \right] m(\log m)^2, \quad (\text{S5.19})$$

$$\sum_x \left| (K_{m,\delta})_{xx} - \frac{\text{tr } K_{m,\delta}}{m} \right| = o_\delta(\text{tr } K_{m,\delta}). \quad (\text{S5.20})$$

Here $o_\delta(1) \rightarrow 0$ as $m \rightarrow \infty$ with δ fixed.

Proof. Use (S5.18) with $h_m = h_{m,\delta}$. The leverage bound becomes $t_m \leq (1+\delta) \log m/\pi + C$. The completion trace is $O(m) + O(m^{1-\delta} \log^2 m) = o_\delta(m \log^2 m)$. The odd-cosine diagonal deviation of $t_m C_m$ is $t_m O(m) = O_\delta(m \log m) = o_\delta(m \log^2 m)$, and the completion has diagonal deviation at most twice its trace. The lower bound $\text{tr } K_{m,\delta} \geq c_0 m \log^2 m$, for an absolute constant $c_0 > 0$, converts both additive remainders into the stated relative remainder. $\square$

Lemma S5.5 (Weighted block and tail extension for singular matrices). *Let $Q \succeq 0$ on a finite-dimensional Hilbert space and suppose*

$$Q \succeq |x_a\rangle\langle x_a|, \quad \text{tr } Q \leq L, \quad u_a = x_a + \gamma_a w, \quad |\gamma_a| \leq 1, \quad \|w\| \leq 1$$

for every label a . If $L > 0$, then

$$\mathcal{K}_L = (1 + L^{-1/2})Q + (1 + \sqrt{L})|w\rangle\langle w|$$

dominates every $|u_a\rangle\langle u_a|$ and satisfies

$$\text{tr } \mathcal{K}_L \leq (\sqrt{L} + 1)^2.$$

If $L = 0$, then $Q = 0$, every $x_a = 0$, and $\mathcal{K}_0 = |w\rangle\langle w|$ is feasible with trace at most one. For the physical right suffix

$$u_{a,c}^+(k) = s(c)c^{k-a}\mathbf{1}_{\{k \geq a\}}$$

and a consecutive block $E = [A, B]$ containing a , the decomposition

$$u_{a,c}^+ = x_{E,a}^+ + c^{B+1-a}w_E^+$$

is exact, where $x_{E,a}^+$ is the restriction to $k \leq B$ and

$$w_E^+(k) = s(c)c^{k-B-1}\mathbf{1}_{\{k \geq B+1\}}.$$

Here $0 \leq c^{B+1-a} \leq 1$ and $\|w_E^+\| \leq 1$. Reflection gives the left suffix decomposition

$$u_{b,c}^- = x_{E,b}^- + c^{b-A+1}w_E^-.$$

Proof. For arbitrary vectors x, y and $\alpha > 0$,

$$\alpha|x\rangle\langle x| + \alpha^{-1}|y\rangle\langle y| - |x\rangle\langle y| - |y\rangle\langle x| = |\sqrt{\alpha}x - \alpha^{-1/2}y\rangle\langle \sqrt{\alpha}x - \alpha^{-1/2}y| \succeq 0.$$

Apply this identity with $y = \gamma_a w$. Since $|w\rangle\langle w| \succeq |\gamma_a w\rangle\langle \gamma_a w|$, choosing $\alpha = L^{-1/2}$ proves feasibility for $L > 0$, and

$$\text{tr } \mathcal{K}_L \leq (1 + L^{-1/2})L + (1 + \sqrt{L}) = (\sqrt{L} + 1)^2.$$

If $L = 0$, positivity and trace zero imply $Q = 0$, hence every $x_a = 0$; the stated conclusion for $L = 0$ follows.

The suffix identity is checked coordinatewise. For $k \geq B+1$, its tail term is $s(c)c^{B+1-a}c^{k-B-1} = s(c)c^{k-a}$, and the coordinates within the block agree by definition. Moreover,

$$\|w_E^+\|^2 = (1 - c^2) \sum_{j=0}^{n-B-1} c^{2j} \leq 1.$$

Reflection about the midpoint of the site set gives the left formula. At $c = 0$, the suffix is a coordinate vector and the decomposition follows by direct inspection; at $c = 1$, every physical suffix is zero. $\square$

Proposition S5.6 (Weighted local Volterra certificate). *Let $K_m^{(0)}$ denote the unweighted certificate supplied by Proposition S5.3, and let*

$$E_{m,c} = \text{diag}(1, c, \dots, c^{m-1}).$$

For $0 < c < 1$, the operator

$$Q_{m,c} = s(c)^2 c^{-2(m-1)} E_{m,c} K_m^{(0)} E_{m,c}$$

dominates the rank-one operator associated with every physical suffix restricted to the block. For all sufficiently large m , it has the explicit upper budget

$$\mathrm{tr} Q_{m,c} \leq \Lambda_{m,c} := \frac{s(c)^2 c^{-2(m-1)}}{\pi^2} m (\log m)^2 \left[1 + C \frac{\log \log m}{\log m} \right].$$

For any sequence of pairs (m_j, c_j) satisfying

$$m_j \rightarrow \infty, \quad 0 < c_j < 1, \quad c_j \rightarrow 1, \quad m_j(1 - c_j) \rightarrow 0, \quad \frac{\log m_j}{\ell(c_j)} \rightarrow 1,$$

one has

$$\Lambda_{m_j, c_j} = m_j p_1(c_j) [1 + o(1)].$$

More precisely, let $\{(m_{n,k}, c_{n,k}) : k \in K_n\}$ be a triangular array with $0 < c_{n,k} < 1$. If

$$\begin{aligned} \inf_{k \in K_n} m_{n,k} &\longrightarrow \infty, & \sup_{k \in K_n} (1 - c_{n,k}) &\longrightarrow 0, \\ \sup_{k \in K_n} m_{n,k} (1 - c_{n,k}) &\longrightarrow 0, & \sup_{k \in K_n} \left| \frac{\log m_{n,k}}{\ell(c_{n,k})} - 1 \right| &\longrightarrow 0, \end{aligned}$$

then the conclusion is uniform in k :

$$\sup_{k \in K_n} \left| \frac{\Lambda_{m_{n,k}, c_{n,k}}}{m_{n,k} p_1(c_{n,k})} - 1 \right| \longrightarrow 0.$$

At $c = 0$, the identity operator is a direct local certificate; at $c = 1$, the zero operator is the direct certificate. Finitely many block sizes not covered by the quantitative estimate are handled by the finite frame sum, with its actual trace used as the budget.

Proof. In local coordinates $j = 1, \dots, m$, let $r_{m,j} = \mathbf{1}_{\{j, \dots, m\}}$. The physical suffix restricted to the block is

$$x_j = s(c) c^{-(j-1)} E_{m,c} r_{m,j}.$$

Congruence of $K_m^{(0)} \succeq |r_{m,j}\rangle \langle r_{m,j}|$ by $E_{m,c}$, together with $c^{-2(m-1)} \geq c^{-2(j-1)}$, proves the domination. Because $0 \leq E_{m,c}^2 \leq I$,

$$\mathrm{tr}(E_{m,c} K_m^{(0)} E_{m,c}) = \mathrm{tr}(K_m^{(0)} E_{m,c}^2) \leq \mathrm{tr} K_m^{(0)}.$$

This proves the displayed budget. Its ratio to $m p_1(c)$ is exactly

$$\frac{\Lambda_{m,c}}{m p_1(c)} = c^{-2(m-1)} \left(\frac{\log m}{\ell(c)} \right)^2 \left[1 + C \frac{\log \log m}{\log m} \right].$$

For the stated sequences this tends to one, since $c_j^{-2(m_j-1)} \rightarrow 1$; the same calculation is uniform for every triangular family on which the four errors in the proposition tend to zero uniformly. The endpoint constructions are defined directly and avoid inverse powers of c . $\square$

Corollary S5.7 (Weighted endpoint-cell certificate). *Fix n and a common overlap $c \in [0, 1]$, and partition the sites into consecutive endpoint blocks $E_1, \dots, E_B$. For each block E_p , let Λ_p be the applicable local trace budget from Proposition S5.6, using its direct endpoint or finite frame budget in the remaining cases. For each block, extend the local operators of Proposition S5.6 by zero and apply Lemma S5.5. This gives right and left suffix certificates $\mathcal{K}_p^+$ and $\mathcal{K}_p^-$ satisfying*

$$\mathrm{tr} \mathcal{K}_p^\pm \leq T_p := (\sqrt{\Lambda_p} + 1)^2.$$

Let P_Δ restrict ordered site pairs to $\Delta_n = \{(u, v) : u < v\}$, and define

$$\mathcal{C}_{p,q} = P_\Delta(\mathcal{K}_p^+ \otimes \mathcal{K}_q^-) P_\Delta^*, \quad \mathcal{K}_{\geq 2,n} = \sum_{p \leq q} \mathcal{C}_{p,q}.$$

Then $\mathcal{K}_{\geq 2,n}$ dominates the physical hull projector of every interval and

$$\text{tr } \mathcal{K}_{\geq 2, n} \leq \sum_{p \leq q} T_p T_q.$$

The operator $\mathcal{K}_{\geq 2, n}$ is the unscaled geometric certificate; the Bayes dual under the uniform prior is $\mathcal{K}_{\geq 2, n}/M_n$.

Proof. If $A \succeq |x\rangle\langle x|$ and $B \succeq |y\rangle\langle y|$, with $A, B \succeq 0$, then

$$A \otimes B - |x \otimes y\rangle\langle x \otimes y| = (A - |x\rangle\langle x|) \otimes B + |x\rangle\langle x| \otimes (B - |y\rangle\langle y|) \succeq 0.$$

Compressing both sides preserves the domination relation. Every interval has its endpoints in a unique cell $p \leq q$. The exact hull identity proves feasibility there, and adding the remaining positive cells cannot weaken it. The diagonal cells are also compressed to $u < v$, never replaced by square cells. Compression decreases the trace of a positive operator, giving the trace bound and the stated normalization. $\square$

S5.3 Triangular compression and the sector bound

Recall that $m = n - 1$ and

$$\Lambda_m = \{(x, y) \in \{1, \dots, m\}^2 : x + y \leq m + 1\}.$$

The excitation pair $s < t$ maps to $(s, n + 1 - t)$, while an interval $[a, b]$, $a < b$, maps to $(a, n + 1 - b)$. The corresponding unweighted containment vector is

$$z_{u,v} = P_{\Lambda_m}(r_{m,u} \otimes r_{m,v}). \quad (\text{S5.21})$$

Lemma S5.8 (Flat-diagonal triangular trace). *If $K_m \succeq 0$, $T_m = \text{tr } K_m$, and $\sum_x |(K_m)_{xx} - T_m/m| \leq \epsilon_m T_m$, then*

$$\text{tr}[P_{\Lambda_m}(K_m \otimes K_m)P_{\Lambda_m}] = \frac{1}{2}T_m^2[1 + O(\epsilon_m + m^{-1})]. \quad (\text{S5.22})$$

Proof. For $k_x = (K_m)_{xx}$ and $\bar{k} = T_m/m$, $\sum_{x,y} |k_x k_y - \bar{k}^2| \leq 2\epsilon_m T_m^2$. Sum over Λ_m and use $|\Lambda_m|/m^2 = 1/2 + O(m^{-1})$. $\square$

With K_m from Proposition S5.3, define

$$\mathcal{L}_n = P_{\Lambda_m}(K_m \otimes K_m)P_{\Lambda_m}. \quad (\text{S5.23})$$

Tensor order and compression give $\mathcal{L}_n \succeq |z_{u,v}\rangle\langle z_{u,v}|$, and

$$\text{tr } \mathcal{L}_n \leq \frac{n^2(\log n)^4}{2\pi^4} \left[1 + C \frac{\log \log n}{\log n} \right]. \quad (\text{S5.24})$$

The exact physical hull vector is $q_n^2 c_n^{|I|} W_n z_I$. Hence

$$\hat{\mathcal{L}}_n = q_n^4 W_n \mathcal{L}_n W_n \quad (\text{S5.25})$$

is feasible for every depth- ≥ 2 sector label, because $c_n^{|I|} \leq 1$. Moreover, $\text{tr } \hat{\mathcal{L}}_n \leq q_n^4 \rho_n^2 \text{tr } \mathcal{L}_n$.

Proposition S5.9 (Critical sector bounds). *Uniformly for $0 \leq \tau_n \leq T$,*

$$M_n P_{\text{opt}}^{(0)} = c_n^2, \quad (\text{S5.26})$$

$$M_n P_{\text{opt}}^{(1)} \leq \frac{4\tau_n}{\pi^2} [1 + O_T((\log n)^{-1})], \quad (\text{S5.27})$$

$$M_n P_{\text{opt}}^{(\geq 2)} \leq \frac{2\tau_n^2}{\pi^4} \left[1 + O_T\left(\frac{\log \log n}{\log n}\right) \right]. \quad (\text{S5.28})$$

Proof. The vacuum vectors are collinear, proving (S5.26). Equation (S5.7), multiplied by s_n^2 , and

$$s_n^2 n (\log n)^2 = 2\tau_n [1 + O_T((n(\log n)^2)^{-1})],$$

prove (S5.27). Equations (S5.24)–(S5.25), together with

$$q_n^2 n (\log n)^2 = 2\tau_n [1 + O_T((n(\log n)^2)^{-1})],$$

and $\rho_n = 1 + O_T((\log n)^{-2})$, prove (S5.28). $\square$

Corollary S5.10 (Critical full optimum upper bound). *Uniformly for $0 \leq \tau_n \leq T$,*

$$\sqrt{M_n P_{\text{opt}}(G_n)} \leq A(\tau_n) + C_T \frac{\log \log n}{\log n}. \quad (\text{S5.29})$$

Proof. Apply Lemma S1.5 to the sectors $0, 1, \geq 2$, and then use Proposition S5.9. $\square$

S6 Uniform matching and hull dominance

S6.1 Inner matching

Proposition S6.1 (Inner matching below the continuum scale). *Suppose*

$$\tau_n = n(1 - c_n)(\log n)^2 \longrightarrow \infty, \quad \lambda_n = n(1 - c_n) = \frac{\tau_n}{(\log n)^2} \longrightarrow 0. \quad (\text{S6.1})$$

Then, for $X \in \{\text{tr}, \text{SRM}, \text{opt}\}$,

$$M_n P_X(G_n) \sim \frac{2\tau_n^2}{\pi^4} \sim A(\tau_n)^2. \quad (\text{S6.2})$$

Proof. Equations (S6.1) imply $c_n^n \rightarrow 1$ and $\rho_n = c_n^{-(n-2)} \rightarrow 1$. Schatten ideal inequalities applied to the exact hull factorization give

$$c_n^n \|Z_n\|_{S_1} \leq \|D_n Z_n W_n\|_{S_1} \leq \rho_n \|Z_n\|_{S_1}. \quad (\text{S6.3})$$

For example, the lower inequality follows by writing $Z_n = D_n^{-1}(D_n Z_n W_n)W_n^{-1}$ and using $\|D_n^{-1}\|_{\text{op}} = c_n^{-n}$ and $\|W_n^{-1}\|_{\text{op}} \leq 1$. Proposition S4.1 and

$$q_n^2 n (\log n)^2 = 2\tau_n [1 + o(1)]$$

therefore give the full physical ≥ 2 -sector trace amplitude

$$\frac{\|A_{\geq 2, n}\|_{S_1}}{\sqrt{M_n}} \sim \frac{\sqrt{2} \tau_n}{\pi^2}. \quad (\text{S6.4})$$

For the reverse sector bound, use Corollary S5.4 with any fixed $0 < \delta < 1$, rather than the moving choice in (S5.15). Its triangular tensor trace is at most

$$\left[\frac{(1 + \delta)^2}{2\pi^4} + o_\delta(1) \right] n^2 (\log n)^4,$$

where $o_\delta(1) \rightarrow 0$ as $n \rightarrow \infty$ with δ fixed. Congruence by $q_n^2 W_n$ gives the explicit physical bound

$$M_n P_{\text{opt}}^{(\geq 2)} \leq q_n^4 \rho_n^2 \left[\frac{(1 + \delta)^2}{2\pi^4} + o_\delta(1) \right] n^2 (\log n)^4.$$

Moreover,

$$\frac{q_n^4 n^2 (\log n)^4}{4\tau_n^2} \longrightarrow 1, \quad \rho_n \longrightarrow 1.$$

Therefore

$$\limsup_n \frac{M_n P_{\text{opt}}^{(\geq 2)}}{2\tau_n^2/\pi^4} \leq (1 + \delta)^2.$$

First take $n \rightarrow \infty$, then $\delta \downarrow 0$, and combine with the subnormalized trace lower bound to obtain equality at leading order. The vacuum sector is $O(1)$, while the cyclic certificate makes the one-excitation sector amplitude $O(\sqrt{\tau_n})$. They are negligible relative to τ_n . Lemma S1.5 and $P_{\text{tr}} \leq P_{\text{SRM}} \leq P_{\text{opt}}$ prove (S6.2). $\square$

To compare with the fixed overlap expression, the complete elliptic-integral form of the one-dimensional functional gives, for $\varepsilon = 1 - c \downarrow 0$,

$$p_1(c) = \frac{2\varepsilon}{\pi^2} \log^2 \frac{8}{\varepsilon} [1 + o(1)], \quad (\text{S6.5})$$

$$p_1(c)^2 = \frac{4\varepsilon^2}{\pi^4} \log^4 \frac{8}{\varepsilon} [1 + o(1)]. \quad (\text{S6.6})$$

Under (S6.1), one has $0 < \varepsilon_n = 1 - c_n = \tau_n/[n(\log n)^2] \rightarrow 0$, and hence

$$M_n p_1(c_n)^2 \sim \frac{2\tau_n^2}{\pi^4} \left[\frac{\log(8n(\log n)^2/\tau_n)}{\log n} \right]^4. \quad (\text{S6.7})$$

More generally, for sequences with $0 < 1 - c_n \rightarrow 0$, the formulas match the inner coefficient precisely when $\log[n(1 - c_n)] = o(\log n)$. This condition holds automatically under (S6.1). Equation (S6.7) compares the two asymptotic formulas directly through the endpoint expansion.

S6.2 Continuum and moving outer regimes

When $\lambda_n = n(1 - c_n)$ stays away from zero, the physical length and hull weights are no longer close to the identity. We therefore retain them within each block through the exact hull coordinates, local block and tail certificates, and matched rectangular pinching instead of using (S6.3).

Lemma S6.2 (Matched rectangular pinching). *Let $A : \mathbb{C}^d \rightarrow \mathbb{C}^r$. Let $\{P_j\}_{j=1}^J$ and $\{Q_j\}_{j=1}^J$ be pairwise orthogonal families of row and column projections. Then*

$$\left\| \sum_{j=1}^J P_j A Q_j \right\|_{S_1} \leq \|A\|_{S_1}.$$

Proof. Use the Hermitian dilation

$$\mathcal{H}(A) = \begin{pmatrix} 0 & A \\ A^* & 0 \end{pmatrix}, \quad \|\mathcal{H}(A)\|_{S_1} = 2\|A\|_{S_1}.$$

The projections $R_j = P_j \oplus Q_j$, together with the complementary projection, form an orthogonal resolution of the dilation space. Ordinary block pinching is contractive in the Schatten 1-norm, and compression to $\sum_j R_j$ leaves the dilation of $\sum_j P_j A Q_j$. Dividing the resulting inequality by two proves the claim. This argument is specific to matched row and column blocks; arbitrary entry deletion need not be contractive. $\square$

Proposition S6.3 (Compact continuum interface). *Let $K = [\lambda_-, \lambda_+] \Subset (0, \infty)$ and $c_{n,\lambda} = 1 - \lambda/n$. For $X \in \{\text{tr}, \text{SRM}, \text{opt}\}$,*

$$\sup_{\lambda \in K} \left| \frac{P_X(G_n(c_{n,\lambda}))}{p_1(c_{n,\lambda})^2} - 1 \right| \rightarrow 0.$$

Equivalently,

$$\sup_{\lambda \in K} \left| \frac{M_n P_X(G_n(c_{n,\lambda}))}{(\log n)^4} - \frac{2\lambda^2}{\pi^4} \right| \rightarrow 0.$$

Proof. Write $c = 1 - \lambda/n$, $s = s(c)$, and $B = \lceil \ell(c) \rceil$. Partition the sites into B balanced consecutive blocks of sizes $m_p \in \{\lfloor n/B \rfloor, \lceil n/B \rceil\}$. Lemma S1.1 gives, uniformly for $\lambda \in K$,

$$\ell(1 - \lambda/n) = \log n + \log(8/\lambda) + o_K(1).$$

Consequently,

$$B \rightarrow \infty, \quad B/n \rightarrow 0, \quad \min_p m_p \rightarrow \infty, \quad \max_p m_p(1 - c) \rightarrow 0, \quad \max_p \left| \frac{\log m_p}{\ell(c)} - 1 \right| \rightarrow 0.$$

The finite-dimensional Volterra law (S4.6) and the exact similarity

$$V_m(c) = E_{m,c}^{-1} U_m E_{m,c}$$

therefore imply, uniformly over all blocks,

$$s \|V_{m_p}(c)\|_{S_1} = m_p \sqrt{p_1(c)} [1 + o_K(1)].$$

Indeed, the two ideal-norm comparisons differ from one by at most $\exp\{O(m_p(1 - c))\}$, and $\sqrt{p_1(c)} = s\ell(c)/\pi$.

For each $p < q$, retain rows with endpoints in $E_p \times E_q$ and strict hull columns in the same rectangle. By (S1.10), the retained matrix is exactly

$$s^2 V_{m_p}(c) \otimes V_{m_q}(c)^\top.$$

Lemma S6.2 and tensor multiplicativity give

$$\|A_{\geq 2,n}(c)\|_{S_1} \geq p_1(c) [1 - o_K(1)] \sum_{p < q} m_p m_q.$$

The restriction to block pairs $p < q$ gives the exact identity

$$\sum_{p < q} m_p m_q = \frac{1}{2} \left(n^2 - \sum_p m_p^2 \right) = M_n [1 - o_K(1)],$$

because $\sum_p m_p^2 \leq n(n/B + 1)$. Thus

$$\|A_{\geq 2,n}(c)\|_{S_1} \geq M_n p_1(c) [1 - o_K(1)].$$

The full Gram matrix dominates $A_{\geq 2,n} A_{\geq 2,n}^*$. Operator monotonicity of the square root, followed by the full external normalization $1/M_n$, yields

$$P_{\text{tr}}(G_n(c)) \geq \left(\frac{\|A_{\geq 2,n}(c)\|_{S_1}}{M_n} \right)^2 \geq p_1(c)^2 [1 - o_K(1)].$$

For the matching upper bound, Proposition S5.6 gives local budgets

$$\Lambda_{p,n} = m_p p_1(c) [1 + o_K(1)].$$

Their minimum diverges: uniformly on K , $m_p p_1(c) \asymp_K \ell(c) \rightarrow \infty$. Lemma S5.5 therefore gives full one-sided suffix certificates with

$$T_{p,n} := (\sqrt{\Lambda_{p,n}} + 1)^2 = m_p p_1(c) [1 + o_K(1)].$$

Corollary S5.7, including all diagonal cells, supplies an unscaled geometric hull certificate satisfying

$$\text{tr } \mathcal{K}_{\geq 2,n} \leq \sum_{p \leq q} T_{p,n} T_{q,n} \leq M_n p_1(c)^2 [1 + o_K(1)].$$

Dividing this geometric certificate by M_n yields $P_{\text{opt}}^{(\geq 2)} \leq p_1(c)^2 [1 + o_K(1)]$.

The vacuum sector is exact: $P_{\text{opt}}^{(0)} = c^2/M_n$. The cyclic certificate (S5.7) gives

$$\kappa_{1,n} \leq Cs^2n(\log n)^2.$$

The compact scalar normalization then yields, uniformly on K ,

$$\frac{\sqrt{P_{\text{opt}}^{(0)}}}{p_1(c)} = O_K((\log n)^{-2}), \quad \frac{\sqrt{P_{\text{opt}}^{(1)}}}{p_1(c)} = O_K((\log n)^{-1}).$$

Lemma S1.5, the hull certificate, and $P_{\text{tr}} \leq P_{\text{SRM}} \leq P_{\text{opt}}$ squeeze all three relative ratios to one.

Finally, the exact scalar identity

$$\frac{M_n p_1(1 - \lambda/n)^2}{(\log n)^4} = \frac{\lambda^2}{2\pi^4} \left(1 + \frac{1}{n}\right) \left(2 - \frac{\lambda}{n}\right)^2 \left(\frac{\ell(1 - \lambda/n)}{\log n}\right)^4$$

converges uniformly to $2\lambda^2/\pi^4$, and is uniformly bounded away from zero on K . This proves the scaled formulation. $\square$

The table records the normalization conventions used here and below.

Object	Constraint or definition	Prior status
$A_{\geq 2,n}$	Exact physical strict-hull state matrix; all M_n rows, including zero singleton rows	no prior
$\mathcal{K}_{\geq 2,n}$	$\mathcal{K}_{\geq 2,n} \succeq \phi_I^{(\geq 2)}\rangle\langle\phi_I^{(\geq 2)} $ for every physical interval	geometric, unscaled
$\mathcal{K}_{\geq 2,n}/M_n$	Feasible Bayes dual for the hull sector	external $1/M_n$
$P_{\text{opt}}^{(k)}$	$\kappa_{k,n}/M_n$, for $k \in \{0, 1, \geq 2\}$	same $1/M_n$ in every sector
$P_{\text{tr}}(G_n)$	$(\text{tr} \sqrt{G_n}/M_n)^2$ for the full physical Gram	full ensemble

Lemma S6.4 (Adaptive macroblock parameters). *Suppose $c_n \rightarrow 1$, $c_n < 1$ eventually, and $h_n = np_1(c_n) \rightarrow \infty$. Set*

$$B_n = \left\lceil \sqrt{h_n} + \lambda_n \sqrt{\ell_n} \right\rceil, \quad m_n = \left\lfloor \frac{n}{B_n} \right\rfloor.$$

Then

$$B_n \rightarrow \infty, \quad \frac{B_n}{h_n} \rightarrow 0, \quad \frac{B_n}{n} \rightarrow 0, \quad \frac{\lambda_n}{B_n} \rightarrow 0,$$

and a balanced B_n -block partition exists eventually. Its minimum block size satisfies

$$m_n(1 - c_n) \rightarrow 0, \quad m_n p_1(c_n) \rightarrow \infty, \quad \frac{\log m_n}{\ell_n} \rightarrow 1.$$

No monotonicity of c_n or λ_n is required.

Proof. Cauchy–Schwarz and the Poisson-kernel integral give $J(c)^2 \leq (1 - c^2)^{-1}$, hence $p_1(c) \leq 1$ and $h_n \leq n$. The exact identity

$$h_n = \frac{\lambda_n(1 + c_n)}{\pi^2} \ell_n^2$$

and the definition of B_n give

$$\frac{B_n}{h_n} \leq \frac{1}{\sqrt{h_n}} + \frac{\pi^2}{(1 + c_n)\ell_n^{3/2}} + \frac{1}{h_n} \rightarrow 0.$$

Also $B_n \geq \sqrt{h_n} \rightarrow \infty$, $\lambda_n/B_n \leq \ell_n^{-1/2} \rightarrow 0$, and $B_n/n \leq B_n/h_n \rightarrow 0$. Thus $B_n \leq n$ eventually and

$$m_n = \frac{n}{B_n} [1 + O(B_n/n)].$$

Multiplication by $1 - c_n = \lambda_n/n$ and by $p_1(c_n) = h_n/n$ proves the first two block limits. It remains to match logarithms. Lemma S1.1 gives

$$\ell_n = \log n - \log \lambda_n + \log 8 + o(1).$$

On indices with $\lambda_n \leq 1$, the identity for h_n and $h_n \rightarrow \infty$ imply $-\log \lambda_n \leq 2 \log \ell_n$ eventually. Hence $\ell_n = \log n + O(\log \ell_n)$, while the block definition gives $\log B_n = O(\log \ell_n)$. On indices with $\lambda_n \geq 1$,

$$\lambda_n \sqrt{\ell_n} \leq B_n \leq C \lambda_n \ell_n$$

eventually, so $\log B_n = \log \lambda_n + O(\log \ell_n)$. In both cases,

$$\log(n/B_n) = \ell_n + O(\log \ell_n).$$

The floor changes this by $o(1)$, and $\ell_n \rightarrow \infty$, proving $\log m_n/\ell_n \rightarrow 1$. Together, the two cases cover every index, even when the sequence oscillates. $\square$

Lemma S6.5 (Adaptive strict-hull squeeze). *Suppose $c_n \rightarrow 1$, $c_n < 1$ eventually, and $\lambda_n = n(1 - c_n) \rightarrow \infty$. The identity $h_n = \lambda_n(1 + c_n)\ell_n^2/\pi^2$ gives $h_n \rightarrow \infty$. Use the balanced partition with B_n blocks and minimum block size m_n from Lemma S6.4, and define*

$$\varepsilon_n^{(\geq 2)} = \frac{\lambda_n}{B_n} + \frac{\log \log m_n}{\log m_n} + \left| \frac{\log m_n}{\ell_n} - 1 \right| + \sqrt{\frac{B_n}{h_n}} + \frac{1}{B_n}.$$

There is an absolute constant $C > 0$ such that, for all sufficiently large n , the exact strict-hull state matrix and the certificate constructed in Corollary S5.7 satisfy

$$\|A_{\geq 2,n}(c_n)\|_{S_1} \geq M_n p_1(c_n) (1 - E_n^{(\geq 2)}), \quad (\text{S6.8})$$

$$\text{tr } \mathcal{K}_{\geq 2,n} \leq M_n p_1(c_n)^2 (1 + E_n^{(\geq 2)}), \quad (\text{S6.9})$$

where $E_n^{(\geq 2)} = C \varepsilon_n^{(\geq 2)} \rightarrow 0$.

Proof. For each pair of distinct blocks, the exact weighted-hull identity (S1.10) and Lemma S6.2 retain the tensor product of the two local weighted Volterra factors. The similarity comparison costs $O(m_n(1 - c_n)) = O(\lambda_n/B_n)$; the finite Volterra trace law costs $O(1/\log m_n)$; and replacing $\log m_n$ by ℓ_n costs $O(|\log m_n/\ell_n - 1|)$. Finally,

$$\sum_{p < q} m_{p,n} m_{q,n} = \frac{1}{2} \left(n^2 - \sum_p m_{p,n}^2 \right) = M_n [1 - O(B_n^{-1})].$$

These estimates prove (S6.8), after enlarging a single absolute constant C ; the term $1/\log m_n$ is absorbed by $\log \log m_n/\log m_n$ for large n .

For the upper bound, Proposition S5.6 gives, uniformly for the two possible block sizes,

$$\frac{\Lambda_{p,n}}{m_{p,n} p_1(c_n)} = 1 + O \left(\frac{\lambda_n}{B_n} + \frac{\log \log m_n}{\log m_n} + \left| \frac{\log m_n}{\ell_n} - 1 \right| \right).$$

Lemma S5.5 changes the local budget to $T_{p,n} = (\sqrt{\Lambda_{p,n}} + 1)^2$. Since $\min_p \Lambda_{p,n} \asymp h_n/B_n \rightarrow \infty$, its relative cost is $O(\sqrt{B_n/h_n})$. Summing the endpoint-cell certificates of Corollary S5.7 over $p \leq q$ adds only $O(B_n^{-1})$, because

$$\sum_{p \leq q} m_{p,n} m_{q,n} = \frac{1}{2} \left(n^2 + \sum_p m_{p,n}^2 \right) = M_n [1 + O(B_n^{-1})].$$

This proves (S6.9). Every term in $\varepsilon_n^{(\geq 2)}$ tends to zero by Lemma S6.4 and its logarithmic estimate, so $E_n^{(\geq 2)} \rightarrow 0$. $\square$

Proposition S6.6 (Moving outer regime). *If $c_n \rightarrow 1$, $c_n < 1$ eventually, and $\lambda_n = n(1 - c_n) \rightarrow \infty$, then, for $X \in \{\text{tr}, \text{SRM}, \text{opt}\}$,*

$$\frac{P_X(G_n(c_n))}{p_1(c_n)^2} \rightarrow 1.$$

Proof. The identity $h_n = \lambda_n(1 + c_n)\ell_n^2/\pi^2$ gives $h_n \rightarrow \infty$. Lemma S6.5 therefore supplies both strict-hull bounds. In particular, $P_{\text{tr}}(G_n) \geq p_1(c_n)^2[1 - o(1)]$, while the Bayes-normalized hull certificate gives $P_{\text{opt}}^{(\geq 2)} \leq p_1(c_n)^2[1 + o(1)]$.

The vacuum relative amplitude is $O(h_n^{-1})$. For one excitation, (S5.7) gives

$$\frac{\sqrt{P_{\text{opt}}^{(1)}}}{p_1(c_n)} = O\left(\frac{\log n}{\sqrt{\lambda_n}\ell_n^2}\right) = o(1).$$

For the last implication, put $q_n^{\text{sc}} = n/\lambda_n = (1 - c_n)^{-1} \rightarrow \infty$. Then $\ell_n = \log q_n^{\text{sc}} + O(1)$, while $\log n = \log \lambda_n + \log q_n^{\text{sc}}$. Both $\log \lambda_n/\sqrt{\lambda_n}$ and $(\sqrt{\lambda_n} \log q_n^{\text{sc}})^{-1}$ tend to zero. Define the remaining sector errors by

$$E_n^{(1)} = \frac{\log n}{\sqrt{\lambda_n}\ell_n^2}, \quad E_n^{(0)} = \frac{1}{h_n}, \quad \mathcal{E}_n = E_n^{(\geq 2)} + E_n^{(1)} + E_n^{(0)}.$$

The preceding estimates show that $\mathcal{E}_n \rightarrow 0$. Lemma S1.5 and $P_{\text{tr}} \leq P_{\text{SRM}} \leq P_{\text{opt}}$ now complete the squeeze. Thus $\mathcal{E}_n$ is also a sufficient control, up to an absolute multiplicative constant, for the relative amplitude error. $\square$

S6.3 Unified hull dominance criterion

Theorem S6.7 (Unified hull dominance criterion). *Suppose*

$$c_n \rightarrow 1, \quad c_n < 1 \text{ eventually}, \quad h_n = np_1(c_n) \rightarrow \infty.$$

For every such sequence, simultaneously for $X \in \{\text{tr}, \text{SRM}, \text{opt}\}$,

$$\frac{P_X(G_n(c_n))}{p_1(c_n)^2} \rightarrow 1.$$

Proof. Fix an arbitrary subsequence and pass to a further subsequence on which $\lambda_n = n(1 - c_n)$ converges in the compactified half-line $[0, \infty]$.

If $\lambda_n \rightarrow 0$, then the exact identity

$$\lambda_n \ell_n^2 = \frac{\pi^2 h_n}{1 + c_n} \rightarrow \infty$$

implies $-\log \lambda_n \leq 2 \log \ell_n$ eventually. Bootstrapping this bound in $\ell_n = \log n - \log \lambda_n + \log 8 + o(1)$ gives $\ell_n/\log n \rightarrow 1$. Therefore

$$\tau_n = \lambda_n (\log n)^2 = \frac{\pi^2 h_n}{(1 + c_n)(\ell_n/\log n)^2} \rightarrow \infty.$$

The hypotheses of Proposition S6.1 now hold. Its normalization agrees with the present one because

$$\frac{M_n p_1(c_n)^2}{2\tau_n^2/\pi^4} = \frac{M_n}{n^2} \frac{(1 + c_n)^2}{2} \left(\frac{\ell_n}{\log n}\right)^4 \rightarrow 1.$$

If $\lambda_n \rightarrow \lambda_* \in (0, \infty)$, choose one compact $K \Subset (0, \infty)$ containing the values eventually and evaluate the uniform bound in Proposition S6.3 at the moving point λ_n . If $\lambda_n \rightarrow \infty$, apply Proposition S6.6.

To apply the inner and outer results along a sparse subsequence, leave the selected values unchanged. At all sufficiently large omitted dimensions, set $\lambda_n = 1/\log n$ in the inner case or $\lambda_n = \sqrt{n}$ in the outer case, and define

$$c_n = 1 - \frac{\lambda_n}{n}.$$

Choose any values in $[0, 1)$ on the remaining finite prefix. In both constructions $0 \leq c_n < 1$ eventually. The extension preserves the selected Gram matrices and satisfies the corresponding regime assumptions. The continuum case requires no extension because Proposition S6.3 is uniform on compact parameter intervals.

Thus every subsequence has a further subsequence on which the displayed ratio tends to one. The standard subsequence criterion proves the full limit. $\square$

S7 Optional labelwise SRM refinement

This optional refinement resolves the SRM contribution label by label. The global critical and hull dominance conclusions depend on Sections S4–S6; the present section adds mean square control of the individual square root diagonals.

Theorem S7.1 (Physical local law). *For every $T < \infty$, with $c_n(t) = 1 - t/[n(\log n)^2]$,*

$$\sup_{0 \leq t \leq T} \frac{1}{M_n} \sum_{I \in \mathcal{I}_n} \left| \sqrt{M_n}(\sqrt{G_n(c_n(t))})_{I,I} - A(t) \right|^2 \longrightarrow 0. \quad (\text{S7.1})$$

The proof combines the retained orthogonality, incidence-trace, and odd-cosine estimates established earlier.

S7.1 One- and two-excitation local laws

We use a finite-codimension estimate when the three excitation strata are made row-orthogonal. Let $(U_m)_{ij} = \mathbf{1}_{\{i \leq j\}}$.

Lemma S7.2 (Finite-codimension Volterra stability). *Let Q_m be an orthogonal projection with $\text{rank}(I - Q_m) \leq r$, where r is fixed, and put*

$$h_m^{(Q)}(j) = \left[(U_m Q_m U_m^*)^{1/2} \right]_{jj}.$$

Uniformly over all such projections,

$$\frac{1}{m} \sum_{j=1}^m \left(\frac{h_m^{(Q)}(j)}{\log m} - \frac{1}{\pi} \right)^2 \longrightarrow 0, \quad \max_j h_m^{(Q)}(j) \leq \frac{2}{\pi} \log m + O(1). \quad (\text{S7.2})$$

Proof. Write $h_m = h_m^{(I)}$. Operator monotonicity gives $0 \leq h_m^{(Q)}(j) \leq h_m(j)$, while

$$\begin{aligned} 0 &\leq \|U_m\|_{S_1} - \|U_m Q_m\|_{S_1} \\ &\leq \|U_m(I - Q_m)\|_{S_1} \leq \sqrt{r} \|U_m\|_{S_2} = O_r(m). \end{aligned}$$

Thus the nonnegative deficit $(h_m - h_m^{(Q)})/\log m$ has empirical mean $O_r((\log m)^{-1})$. The odd-cosine expansion in Lemma S5.2 gives

$$\frac{1}{m} \sum_j \left(\frac{h_m(j)}{\log m} - \frac{1}{\pi} \right)^2 \rightarrow 0, \quad \max_j h_m(j) \leq \frac{2}{\pi} \log m + O(1).$$

The normalized deficit is therefore uniformly bounded; its empirical second moment is bounded by a constant times its empirical first moment. This proves both assertions. $\square$

Let $H_{1,n} = (B_{1,n} B_{1,n}^*)^{1/2}$ and $h_{1,n}(I) = (H_{1,n})_{I,I}$.

Lemma S7.3 (One-excitation empirical law). *With $N = n + 1$,*

$$\frac{1}{M_n} \sum_{I \in \mathcal{I}_n} \left[\frac{\sqrt{N}}{\log N} h_{1,n}(I) - \frac{2}{\pi} \right]^2 \longrightarrow 0. \quad (\text{S7.3})$$

Moreover,

$$\max_{I \in \mathcal{I}_n} h_{1,n}(I) \leq C \frac{\log N}{\sqrt{N}}. \quad (\text{S7.4})$$

Proof. The Dirichlet eigenbasis in (S4.1) gives, for $I = [a, b]$ and $L = b - a + 1$,

$$h_{1,n}([a, b]) = \frac{4}{N^{3/2}} \sum_{k=1}^{N-1} \frac{\sin^2(L\theta_k/2) \sin^2[(a+b)\theta_k/2]}{\sin(\theta_k/2)}, \quad \theta_k = \frac{\pi k}{N}. \quad (\text{S7.5})$$

Set $T_N(m) = \sum_{k=1}^{N-1} \cos(m\theta_k) / \sin(\theta_k/2)$. Expanding the product of sine squares in (S7.5) gives

$$\frac{1}{4}T_N(0) - \frac{1}{4}T_N(L) - \frac{1}{4}T_N(a+b) + \frac{1}{8}T_N(2a-1) + \frac{1}{8}T_N(2b+1).$$

The zero frequency is $T_N(0) = (2N/\pi) \log N + O(N)$. Discrete cosine orthogonality gives

$$\sum_{m=0}^{2N-1} T_N(m)^2 = N \sum_{k=1}^{N-1} \csc^2\left(\frac{\pi k}{2N}\right) = \frac{2N(N^2-1)}{3}. \quad (\text{S7.6})$$

Each of the four nonzero frequency maps has at most n interval preimages. Equation (S7.6) therefore makes their normalized empirical square contribution $O((\log N)^{-2})$, proving (S7.3). Since $|T_N(m)| \leq T_N(0)$ for every integer m , the same exact diagonal formula also proves (S7.4). $\square$

If $H_{1,n}^{(D)} = (D_n B_{1,n} B_{1,n}^* D_n)^{1/2}$ and $\epsilon_n = 1 - c_n^n$, the rectangular polar formula gives the pointwise comparison

$$(1 - \epsilon_n)^2 h_{1,n}(I) \leq (H_{1,n}^{(D)})_{I,I} \leq (1 - \epsilon_n)^{-1} h_{1,n}(I). \quad (\text{S7.7})$$

Indeed, compare $B_{1,n}^* D_n^2 B_{1,n}$ with $B_{1,n}^* B_{1,n}$, invert, and apply square root monotonicity before taking the polar diagonal. In a compact critical window, $\epsilon_n = O_T((\log n)^{-2})$. Thus

$$\sup_{0 \leq t \leq T} \frac{1}{M_n} \sum_I \left| \sqrt{M_n} q_n(H_{1,n}^{(D)})_{I,I} - \frac{2\sqrt{t}}{\pi} \right|^2 \longrightarrow 0. \quad (\text{S7.8})$$

For the two-endpoint law, restrict Z_n to its nonzero proper-interval rows and write $H_{2,n} = (Z_n Z_n^*)^{1/2}$. Map (a, b) to $(u, v) = (a, n+1-b)$ in Λ_{n-1} .

Lemma S7.4 (Two-excitation empirical law). *For $N_n = \binom{n}{2}$,*

$$\frac{1}{N_n} \sum_{a < b} \left[\frac{(H_{2,n})_{(a,b),(a,b)}}{(\log n)^2} - \frac{1}{\pi^2} \right]^2 \longrightarrow 0. \quad (\text{S7.9})$$

Proof. The interval-poset zeta matrix is invertible; its inverse is the mixed first-difference operator

$$x(a, b) = y(a, b) - y(a+1, b) - y(a, b-1) + y(a+1, b-1).$$

Let $D_m = U_m^{-1}$, $m = n-1$, and let V_m extend functions on Λ_m by zero to the full square. Forward differences preserve the triangular support, so $(D_m \otimes D_m)V_m = V_m Z_n^{-1}$. Since $x \mapsto x^{-1/2}$ is operator convex, compression Jensen gives the pointwise comparison

$$(H_{2,n})_{(a,b),(a,b)} \leq d_{m,u} d_{m,v}, \quad d_{m,x} = ((U_m U_m^*)^{1/2})_{xx}. \quad (\text{S7.10})$$

The sine expansion used in Lemma S5.2 gives

$$\frac{1}{m} \sum_x \left(\frac{d_{m,x}}{\log m} - \frac{1}{\pi} \right)^2 \rightarrow 0, \quad \max_x d_{m,x} \leq \frac{2}{\pi} \log m + O(1).$$

The triangular restriction preserves this empirical convergence: for any sequence x_u and constant a ,

$$\frac{1}{N_n} \sum_{u+v \leq m+1} |x_u - a|^2 \leq \frac{2}{m} \sum_u |x_u - a|^2.$$

The product bound and uniform estimate above therefore make the right-hand side of (S7.10) converge to $1/\pi^2$ in empirical L^2 over the triangle. Proposition S4.1 says the left-hand side has the same empirical mean. The normalized deficit is nonnegative and uniformly bounded. If $\Delta_{a,b}$ denotes this deficit, then

$$\frac{1}{N_n} \sum_{a < b} \Delta_{a,b}^2 \leq \left(\max_{a < b} \Delta_{a,b} \right) \frac{1}{N_n} \sum_{a < b} \Delta_{a,b} \rightarrow 0.$$

The triangle inequality in empirical L^2 now proves (S7.9). $\square$

The same polar comparison as in (S7.7), followed by the hull order (S1.12), transfers (S7.9) to all depths ≥ 2 :

$$\sup_{0 \leq t \leq T} \frac{1}{M_n} \sum_I \left| \sqrt{M_n} (\sqrt{A_{\geq 2,n} A_{\geq 2,n}^*})_{I,I} - \frac{\sqrt{2}t}{\pi^2} \right|^2 \rightarrow 0. \quad (\text{S7.11})$$

S7.2 A Kac–Murdock–Szegő diagonal bound

Let $T_m(c) = (c^{|j-k|})_{j,k=1}^m$.

Lemma S7.5 (Uniform KMS square root diagonal). *Fix $C_0 < \infty$ and $\kappa > 0$. There is $C = C(C_0, \kappa)$ such that, whenever*

$$0 < c \leq 1, \quad m(1-c) \leq C_0, \quad c^m \geq \kappa,$$

one has

$$\max_j (\sqrt{T_m(c)})_{jj} \leq C \left[m^{-1/2} + \sqrt{1-c} \log(m+1) \right]. \quad (\text{S7.12})$$

For $c = c_n(t) = 1 - t/[n(\log n)^2]$, uniformly in $0 \leq t \leq T$ and $1 \leq m \leq n$,

$$\max_j (\sqrt{T_m(c_n(t))})_{jj} \leq \frac{C_T}{\sqrt{m}}. \quad (\text{S7.13})$$

Proof. The cases $c = 1$ and $m = 1$ are immediate. For $0 < c < 1$,

$$T_m(c)^{-1} = \frac{1}{1-c^2} \begin{pmatrix} 1 & -c & & \\ -c & 1+c^2 & -c & \\ & \ddots & \ddots & \ddots \\ & & -c & 1 \end{pmatrix}. \quad (\text{S7.14})$$

An eigenvector of the tridiagonal numerator has the form

$$v_j = A[\sin(j\theta) - c \sin((j-1)\theta)], \quad 0 < \theta < \pi, \quad (\text{S7.15})$$

where θ satisfies the right endpoint condition

$$\sin((m+1)\theta) - 2c \sin(m\theta) + c^2 \sin((m-1)\theta) = 0. \quad (\text{S7.16})$$

Indeed, the left endpoint equation is automatic for (S7.15), while the right endpoint equation is equivalent to $v_{m+1} = cv_m$ after extending the same formula to $j = m + 1$. Every nonzero sequence $y_j = \sin(j\theta + \phi)$, $1 \leq j \leq m$, satisfies the uniform sampling inequality

$$\max_j |y_j|^2 \leq \frac{C_s}{m} \sum_{j=1}^m |y_j|^2. \quad (\text{S7.17})$$

To verify it, reduce θ to its distance $d \in [0, \pi/2]$ from $\pi\mathbb{Z}$. If $md > 2$, the two eigenvalues of the sampling Gram matrix with rows $(\cos jd, \sin jd)$ are

$$\frac{m}{2} \pm \frac{|\sin(md)|}{2 \sin d};$$

the smaller is at least a fixed positive multiple of m . If $md \leq 2$, write $u = md$ and use the uniformly nondegenerate basis $\cos(ut)$ and $\sin(ut)/u$, with its continuous limit t at $u = 0$. Its normalized Riemann-sum Gram matrix has a uniformly positive smallest eigenvalue for $u \in [0, 2]$. This proves (S7.17), including the finitely many small values of m by compactness. Applied to (S7.15), it gives

$$\max_j |v_j|^2 \leq C/m \quad (\text{S7.18})$$

for every normalized eigenvector.

The standard Cholesky factor of $T_m(c)$ has first column $(1, c, \dots, c^{m-1})^\top$ and lower-right block $\sqrt{1 - c^2} V_{m-1}(c)$, where

$$V_s(c) = D_{c,s} U_s^\top D_{c,s}^{-1}, \quad D_{c,s} = \text{diag}(1, c, \dots, c^{s-1}).$$

The assumption $c^m \geq \kappa$ bounds the condition number of $D_{c,s}$, so $\|V_s(c)\|_{S_1} \leq C \|U_s\|_{S_1}$. Consequently,

$$\text{tr} \sqrt{T_m(c)} \leq \sqrt{m} + C \sqrt{1 - c} m \log(m + 1). \quad (\text{S7.19})$$

The spectral resolution, (S7.18), and (S7.19) prove (S7.12). In the critical family, $c_n(t)^m \geq c_n(t)^n \geq 1/2$ for large n , while

$$\sqrt{1 - c_n(t)} \log(m + 1) \leq \frac{\sqrt{T}}{\sqrt{n}} \leq \frac{\sqrt{T}}{\sqrt{m}}.$$

This proves (S7.13) uniformly on $[0, T]$. $\square$

S7.3 Completion of the local law

Proof of Theorem S7.1. Every column of $B_{1,n}$ is endpoint-additive, because

$$\mathbf{1}_{\{a \leq s \leq b\}} = \mathbf{1}_{\{a \leq s\}} + \mathbf{1}_{\{s \leq b\}} - 1.$$

Lemma S3.1 therefore shows that the three terms in

$$K_n = J_{M_n} + q_n^2 B_{1,n} Q_{1,n} B_{1,n}^* + q_n^4 Z_n Q_{2,n} Z_n^* \quad (\text{S7.20})$$

have mutually orthogonal row ranges and $0 \preceq K_n \preceq \hat{G}_n$. The projection $Q_{1,n}$ removes one direction. Operator monotonicity, Lemma S7.3, and

$$0 \leq \|B_{1,n}\|_{S_1} - \|B_{1,n} Q_{1,n}\|_{S_1} \leq \|B_{1,n}\|_{\text{op}} = O(n^{3/2})$$

give the same empirical law for its retained one-excitation diagonal. In each two-endpoint cell, the factors in (S3.2) lose at most two directions. Lemma S7.2 applies uniformly because every block size tends to infinity and $\log |E_p| / \log n \rightarrow 1$. Because the factors are uniformly bounded, their products preserve empirical L^2 convergence. The omitted same-block cells occupy $O(r_n^{-1})$ of the interval triangle. Hence, with

$$\ell_{n,I} = \sqrt{M_n}(\sqrt{K_n})_{I,I},$$

we obtain, uniformly on every compact parameter interval,

$$\sup_{0 \leq t \leq T} \frac{1}{M_n} \sum_I |\ell_{n,I} - A(t)|^2 \longrightarrow 0, \quad (\text{S7.21})$$

$$\sup_{0 \leq t \leq T} \left| \frac{\text{tr } \sqrt{K_n}}{\sqrt{M_n}} - A(t) \right| \longrightarrow 0. \quad (\text{S7.22})$$

The maximum estimates in Lemmas [S7.3](#) and [S7.2](#), together with the critical factors q_n and q_n^2 , also give $\sup_{n,t,I} \ell_{n,I} < \infty$ on every compact t -window.

It remains to show that the nonnegative diagonal excess of $\sqrt{\widehat{G}_n}$ cannot concentrate. Fix a site x and consider all intervals containing it. Their physical principal Gram matrix is exactly

$$T_x(c_n) \otimes T_{n+1-x}(c_n). \quad (\text{S7.23})$$

Compression Jensen for the operator-concave square root and Lemma [S7.5](#) imply, for every $I \ni x$,

$$(\sqrt{G_n})_{I,I} \leq \frac{C_T}{\sqrt{x(n+1-x)}}. \quad (\text{S7.24})$$

Let $\chi_n(I) = \max_{x \in I} x(n+1-x)$. Choosing the best anchor gives

$$M_n(\sqrt{G_n})_{I,I}^2 \leq C_T \frac{M_n}{\chi_n(I)}. \quad (\text{S7.25})$$

The right-hand side is uniformly integrable under the interval prior. For large K , only the two edge triangles can satisfy $M_n/\chi_n(I) > K$; on the left edge this forces $b < n/K$, and hence

$$\frac{1}{M_n} \sum_I \frac{M_n}{\chi_n(I)} \mathbf{1}_{\{M_n/\chi_n(I) > K\}} \leq 2 \sum_{b < n/K} \frac{1}{n+1-b} \leq \frac{C}{K}. \quad (\text{S7.26})$$

Let $\widehat{\eta}_{n,I} = \sqrt{M_n}(\sqrt{\widehat{G}_n})_{I,I}$. Operator monotonicity, [\(S7.22\)](#), and [\(S4.10\)](#) give

$$e_{n,I} := \widehat{\eta}_{n,I} - \ell_{n,I} \geq 0, \quad \frac{1}{M_n} \sum_I e_{n,I} \rightarrow 0.$$

Put $\alpha_n = c_n^n$ and $\widetilde{G}_n = D_n \widehat{G}_n D_n$. If $c_n < 1$, the interval-zeta minor makes $\widehat{G}_n$ positive definite. For an invertible square factor $A_n A_n^* = \widehat{G}_n$,

$$\alpha_n^2 A_n^* A_n \preceq A_n^* D_n^2 A_n \preceq A_n^* A_n.$$

Order reversal under inversion, square root monotonicity, and the polar formula give the point-wise diagonal inequalities

$$\alpha_n^2 (\sqrt{\widehat{G}_n})_{I,I} \leq (\sqrt{\widetilde{G}_n})_{I,I} \leq \alpha_n^{-1} (\sqrt{\widehat{G}_n})_{I,I}. \quad (\text{S7.27})$$

Indeed,

$$\begin{aligned} \sqrt{\widehat{G}_n} &= A_n (A_n^* A_n)^{-1/2} A_n^*, \\ \sqrt{\widetilde{G}_n} &= D_n A_n (A_n^* D_n^2 A_n)^{-1/2} A_n^* D_n, \end{aligned}$$

and $\alpha_n \leq (D_n)_{I,I} \leq 1$. Since $\widetilde{G}_n \preceq G_n$, equation [\(S7.27\)](#) yields

$$(\sqrt{\widehat{G}_n})_{I,I} \leq \alpha_n^{-2} (\sqrt{G_n})_{I,I}.$$

Thus (S7.25)–(S7.26) transfer uniform integrability to $\widehat{\eta}_{n,I}^2$. If $c_n = 1$, all matrices in this comparison equal J_{M_n} , so the conclusion is direct. Splitting at a fixed level R , using $e^2 \leq Re$ below that level and the uniform-integrability tail above it, yields

$$\frac{1}{M_n} \sum_I e_{n,I}^2 \rightarrow 0. \quad (\text{S7.28})$$

Combining (S7.21) and (S7.28) first proves the local law for $\widehat{G}_n$. Because $\alpha_n = 1 + O_T((\log n)^{-2})$, (S7.27) and the bounded empirical second moments transfer it to $\widetilde{G}_n$. Finally, Corollary S1.3 gives

$$\sqrt{\widetilde{G}_n} \preceq \sqrt{G_n} \preceq \rho_n \sqrt{\widetilde{G}_n}, \quad \rho_n = 1 + O_T((\log n)^{-2}),$$

so the same argument transfers the law to the exact physical matrix. $\square$

S8 Finite size numerical diagnostics

S8.1 Postprocessed floating point SDP endpoints

Raw solver objectives are retained as diagnostics; the postprocessed feasible matrices define the reported bounds. Let ρ_j denote the prior weighted canonical states on their d_{supp} -dimensional support. For a numerical dual matrix Y , set

$$\delta_D = \max_j \max\{0, -\lambda_{\min}(Y - \rho_j)\}$$

and add the recorded roundoff margin $\zeta_D > 0$, scaled to the numerical problem. Then

$$Y_{\text{safe}} = Y + (\delta_D + \zeta_D)I, \quad U_{\text{safe}} = \text{tr } Y_{\text{safe}} \quad (\text{S8.1})$$

is rechecked against every dual PSD constraint. If the rechecked matrix satisfies these constraints exactly, U_{safe} is a feasible upper bound.

For the primal matrices, project each raw M_j onto the PSD cone and put $S = \sum_j M_j^+$. Choose $\alpha \geq \max\{1, \lambda_{\max}(S)\}$, with an outward margin chosen for the numerical scale, and define

$$\begin{aligned} Q_j &= \frac{M_j^+}{\alpha}, \\ R &= I - \sum_k Q_k \succeq 0, \\ \overline{M}_j &= Q_j + \delta_{j1}R. \end{aligned} \quad (\text{S8.2})$$

This PSD projection and normalization give $\overline{M}_j \succeq 0$ and $\sum_j \overline{M}_j = I$ in exact arithmetic. If q is the number of hypotheses, the final positive floor $f_P > 0$, chosen with $qf_P < 1$, is introduced through the uniform-POVM mixture

$$\widehat{M}_j = (1 - qf_P)\overline{M}_j + f_P I. \quad (\text{S8.3})$$

Thus every $\widehat{M}_j$ is positive definite and the family remains complete. The implementation rechecks the resulting POVM. In exact arithmetic, feasibility and weak duality give

$$L_{\text{safe}} = \sum_j \text{tr}(\rho_j \widehat{M}_j), \quad L_{\text{safe}} \leq P_{\text{opt}} \leq U_{\text{safe}}. \quad (\text{S8.4})$$

The implementation uses IEEE double precision and conventional eigensolvers, with no directed rounding, interval arithmetic, or exact rational verification. Accordingly, L_{safe} and U_{safe} are reported as postprocessed floating point endpoints. In exact arithmetic, the sandwich holds whenever the postprocessed matrices are feasible, and the stored residuals record the feasibility checks at working precision.

S8.2 Critical-window physical Gram checks

For each (n, τ) , the diagnostic calculation sets $c = 1 - \tau/[n(\log n)^2]$, constructs the full physical Gram matrix explicitly as a dense matrix, and diagonalizes it. Table 1 reports the scaled trace amplitude, the target $A(\tau)$, the empirical root-mean-square error of the scaled diagonal, and the trace loss of the row-orthogonal retained comparison. The slow finite size drift, especially at $\tau = 4$, is consistent with the logarithmic theorem and cautions against interpreting an initially increasing polar/SRM gap as asymptotic behavior.

Table 1: Dense physical Gram diagnostics. The retained trace gap is $(\text{tr } \sqrt{G} - \text{tr } \sqrt{K})/\sqrt{M_n}$.

τ	n	$A(\tau)$	trace amplitude	local RMS	retained gap
0.25	12	1.354132	1.442930	0.105208	0.136272
0.25	24	1.354132	1.417693	0.084748	0.107393
0.25	32	1.354132	1.410576	0.079250	0.099183
1.00	12	1.779910	1.965364	0.221155	0.410377
1.00	24	1.779910	1.919312	0.186629	0.320009
1.00	32	1.779910	1.905585	0.176503	0.295459
4.00	12	2.846399	3.165847	0.395940	1.490082
4.00	24	2.846399	3.124405	0.388835	1.105722
4.00	32	2.846399	3.107347	0.381736	1.012978

At $n = 32$, the differences between the full square root diagonal and the sum of the sectorwise square root diagonals have the ranges

τ	0.25	1	4
range	$[-0.2175, 0.1660]$	$[-0.5349, 0.4288]$	$[-1.0370, 1.1270]$

Both signs occur at the working precision, providing numerical evidence against coordinatewise square root additivity. The minimum eigenvalue of $\hat{G}_n - K_n$ lay between -2.9×10^{-15} and -2.2×10^{-15} , and the retained row-orthogonality residuals were below 1.8×10^{-13} , consistent with positive semidefiniteness at the working precision.